

Single cell multi-omics analysis of chronic myeloid leukemia links cellular heterogeneity to therapy response

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Abstract

The advent of tyrosine kinase inhibitors (TKIs) as treatment of chronic myeloid leukemia (CML) is a paradigm in molecularly targeted cancer therapy. Nonetheless, TKI insensitive leukemia stem cells (LSCs) persist in most patients even after years of treatment. The sustained presence, heterogeneity and evolvability of LSCs are imperative for disease progression as well as recurrence during treatment-free remission (TFR). However, dynamic changes among LSC sub-populations upon TKI therapy impede their measurement and targeting. Here, we used cellular indexing of transcriptomes and epitopes by sequencing (CITE-seq) to generate high-resolution single cell multiomics maps from CML patients at diagnosis, retrospectively stratified by BCR::ABL1^{IS} (%) following 12 months of TKI therapy as per European LeukemiaNet (ELN) recommendations. Simultaneous measurement of global gene expression profiles together with >40 surface markers from the same cells revealed that each patient harbored a unique composition of stem and progenitor cells at diagnosis demonstrating that cellular heterogeneity is a hallmark of CML. The patients with treatment failure after 12 months of therapy had markedly higher abundance of molecularly defined primitive cells at diagnosis compared to the optimal responders. Furthermore, deconvolution of an independent dataset of CML patient-derived bulk transcriptomes (n=59) into constituent cell populations showed that the proportion of primitive cells versus lineage primed sub-populations significantly connected with the TKI-treatment outcome. The multiomic feature landscape enabled visualization of the primitive fraction as a heterogeneous mixture of molecularly distinct Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁺ LSCs and BCR::ABL1⁻ hematopoietic stem cells (HSCs) in variable ratio across patients and guided their prospective isolation by a combination of CD26 and CD35 cell surface markers. We for the first time show that BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs can be distinctly separated as CD26⁺CD35⁻ and

CD26⁺CD35⁺ respectively. In addition, we found the relative proportion of CD26⁺CD35⁺ HSCs to be higher in optimal responders when compared to treatment failures, at diagnosis as well as following 3 months of TKI therapy, and that the LSC/HSC ratio was increased in patients with prospective treatment failure. Collectively, the patient-specific cellular heterogeneity multiomics maps build a framework towards understanding therapy response and adapting treatment by devising strategies that either extinguish TKI-insensitive LSCs or engage the immune effectors to suppress the residual leukemogenic cells.

eLife assessment

This study presents **fundamental** insights into the heterogeneity of chronic myeloid leukemia (CML) stem cells and their response to tyrosine kinase inhibitor therapy, shedding light on potential mechanisms underlying treatment failure. The study's robust methodology, supported by validation with bulk RNA-seq data and surface marker analysis, provides **compelling** evidence for the identified associations between cellular composition and treatment outcome. These findings contribute to our understanding of CML pathogenesis and may inform the development of more targeted therapeutic strategies.

Introduction

The persistence, burden and evolvability of leukemic stem cells (LSCs) during therapy provide common threads to address the major challenges remaining in chronic myeloid leukemia (CML) (Cortes et al., 2021 [↗](#); Michor et al., 2005 [↗](#)). For example, acquisition of the BCR::ABL1 kinase domain (KD), and additional somatic mutations within LSCs could either necessitate a switch to another tyrosine kinase inhibitor (TKI) or precipitate a blast crisis (Bolton-Gillespie et al., 2013 [↗](#); Giustacchini et al., 2017 [↗](#); Nieborowska-Skorska et al., 2012 [↗](#); O'Hare et al., 2007 [↗](#); Sorel et al., 2004 [↗](#)). Moreover, even in the absence of KD mutations, the LSC burden at diagnosis is linked to cytogenetic and molecular response in chronic phase (Khorashad et al., 2006 [↗](#); Mustjoki et al., 2013 [↗](#)), and their presence is associated with recurrence during treatment-free remission (TFR) (Pagani et al., 2020 [↗](#)), thus compelling life-long treatment in the majority of patients.

The de facto standard for disease monitoring, BCR::ABL1 transcript/gDNA quantification using qPCR, Sanger and/or Next Generation Sequencing (NGS), however, predictably only captures cells positive for BCR::ABL1 expression and translocation and does not inform the number and identity of fully leukemogenic cells (Radich et al., 2019 [↗](#)). This is pertinent given that a combination of cell sorting and BCR::ABL1 quantification suggested that the presence of BCR::ABL1 signal in the stem cell compartment rather than the lymphocytes correlates with TFR loss (Kinstrie et al., 2020 [↗](#); Pagani et al., 2020 [↗](#)). Using hematopoietic stem cell (HSC) specific markers however, is confounded by the observations that in remission, patients in chronic phase restore Philadelphia chromosome-negative (Ph⁻) hematopoiesis (Bergamaschi et al., 1994 [↗](#); Carella et al., 1993 [↗](#); Claxton et al., 1992 [↗](#); Coulombel et al., 1983 [↗](#); Deininger, 2003 [↗](#)), implying the co-existence of BCR::ABL1⁻ HSCs and BCR::ABL1⁺ LSCs. However, because their proportion varies across individuals and treatment stages (Mustjoki et al., 2013 [↗](#)), and both LSCs and HSCs reside within similar immunophenotypic compartment (Lin⁻CD34⁺CD38^{low}) (Eisterer et al., 2005 [↗](#); Petzer et al., 1996 [↗](#)), discrimination between these populations – and thereby reliably estimating LSCs – remains difficult. As a result, understanding CML LSC identity, heterogeneity, and vulnerability to TKI therapy remain outstanding challenges in CML.

We recently implemented BCR::ABL1 targeted single-cell RT-qPCR in combination with index sorting for surface markers to dissect the LSC compartment at diagnosis and following 3 months of TKI-therapy (Warfvinge et al., 2017 [DOI](#)). This demonstrated that at diagnosis, Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁺ cells could be divided into seven molecularly distinct subpopulations, each reflecting unique lineage and cell state signatures. Importantly, the proportions of BCR::ABL1⁺ LSC subpopulations changed dynamically upon therapy and only a few surface markers efficiently discriminated between BCR::ABL1⁺ vs. BCR::ABL1⁻ cells. Of these, despite being generally reduced in BCR::ABL1⁺ cells upon therapy, CD26 was more frequently detected on the subpopulation most persistent during TKI treatment. The substantial treatment-induced changes observed within the stem cell population suggested that BCR::ABL1⁺ LSCs are themselves heterogeneous in terms of sensitivity to TKIs allowing us to define BCR::ABL1⁺Lin⁻CD34⁺CD38^{-/low}CD45RA⁻cKIT⁻CD26⁺ as the most TKI-insensitive cells. Apart from being detected at diagnosis, these cells are strikingly enriched after TKI therapy (Warfvinge et al., 2017 [DOI](#)), a finding since confirmed in a transgenic mouse model (Shah et al., 2023 [DOI](#)), and in clinical observations documenting long-term persistence of CD26⁺ cells in CML patients (Pacelli et al., 2023 [DOI](#)). Together, these findings argue that functional heterogeneity within LSCs cannot be predicted solely by surface markers but is intimately linked to their cell state and gene expression signature, thereby motivating a strategy to simultaneously capture their lineage potency/affiliation, BCR::ABL1 status, and molecular program in addition to the surface markers.

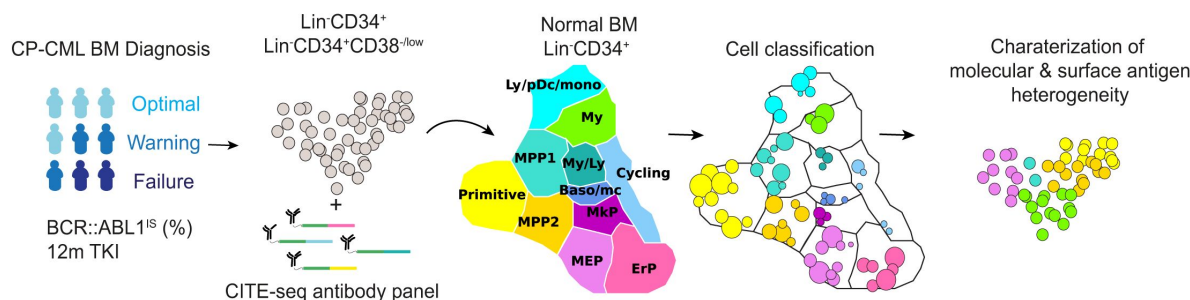
Although recent single cell omics studies have begun to explore the cellular landscape in CML (Giustacchini et al., 2017 [DOI](#); Krishnan et al., 2023 [DOI](#); Patel et al., 2022 [DOI](#); Zhang et al., 2020 [DOI](#)), these have largely been limited to measuring single modalities at a time, and thus, multiomic investigation of BCR::ABL1⁺ LSCs and BCR::ABL1⁻ stem cells from the same patient are currently lacking. Therefore, we used cellular indexing of transcriptomes and epitopes by sequencing (CITE-seq) (Stoeckius et al., 2017 [DOI](#)) to quantify the global gene expression program, and highly multiplexed cell-surface protein profile simultaneously within the same cells from nine CML patients. The single cell multiomics maps enabled us to visualize the cellular heterogeneity of each patient by delineating the abundance of molecularly defined leukemic stem versus lineage progenitors, and link cell composition at diagnosis to TKI therapy response. Moreover, the maps highlighted the co-existence of molecularly distinct BCR::ABL1⁺ vs. BCR::ABL1⁻ stem cells, guiding their prospective isolation by novel combinations of CD26 and CD35 markers, dissection of imbedded molecular programs, division kinetics and changes in heterogeneity following TKI therapy.

Results

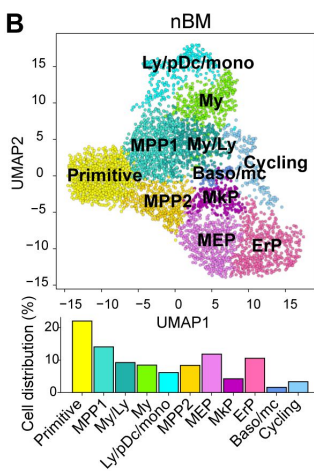
Single-cell multiomic CITE-seq analysis defines the heterogeneity of CML stem cell fractions

To characterize the multimodal heterogeneity within the Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} CML bone marrow (BM) compartment at diagnosis and estimate how the heterogeneity contributes to therapy response, we applied CITE-seq on nine patient retrospectively stratified according to molecular response following 12 months of treatment. To allow for accurate specification of LSPC heterogeneity without confounding effects generated from cell-type-irrelevant transcriptional signals related to BCR-ABL transformation per se, or inter patient variability due to batch effects, we used Scarf projection (Dhapola et al., 2022 [DOI](#)) to map each individual CML cell to a Lin⁻CD34⁺ healthy bone marrow reference with a predefined heterogeneity. Each CML cell was then annotated by label transfer according to the annotation of the reference cells with highest molecular similarity (Fig. 1A [DOI](#)).

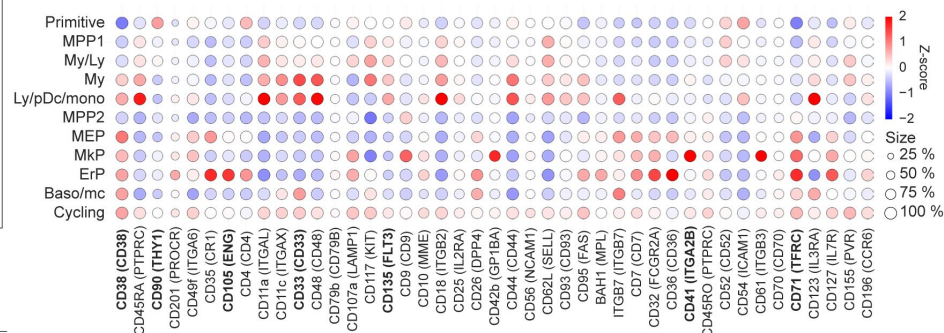
A



B



C



D

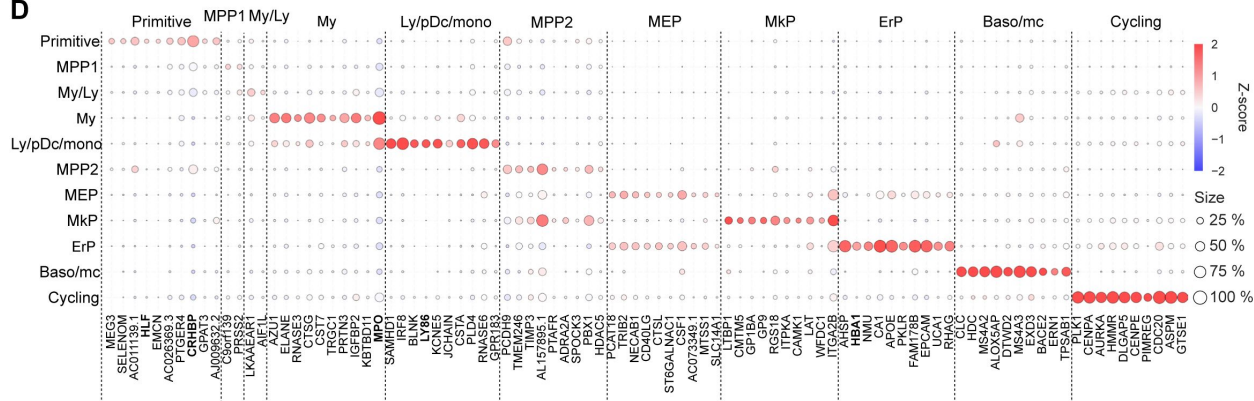


Figure 1.

Single cell multiomics analysis of CML and normal bone marrow (nBM) by CITE-seq

(A) CML LSC and progenitor populations from nine CML BM samples at diagnosis were FACS sorted and subjected to CITE-seq. CML heterogeneity was defined by label transfer through cell projection onto a reference UMAP of age-matched nBM from two donors (Sommarin et al., 2021). Diagnostic samples were retrospectively stratified by BCR::ABL1 International Scale (IS) % after 12 months of TKI treatment, according to the European LeukemiaNet recommendations.

(B) UMAP embedding of 4,696 Lin[−]CD34⁺ nBM cells (n = 2) with 11 clusters identified by Leiden clustering; annotated by marker genes and ADT expression (HSC = hematopoietic stem cell; MPP = multipotent progenitor population 1; My/Ly = myeloid, lymphoid progenitors; My = myeloid progenitors; Ly/pDC/mono = lymphoid, dendritic, monocytic progenitors; MPP2 = multipotent progenitor population 2; MEP = megakaryocytic erythroid progenitors; MkP = megakaryocytic erythroid progenitors; ErP = erythroid progenitors; Baso/mc = basophilic, mast cell progenitors; Cycling = Cycling progenitors). The bar plot below shows the percentage of cells per cluster.

(C) Dot plot showing the ADT expression within clusters. The color indicates mean ADT expression (red = high expression, blue = low expression); dot size represents the fraction of cells with expression.

(D) Dot plot showing mRNA expression of the top 10 marker genes per cluster. The color indicates mean RNA expression (red = high expression, blue = low expression); dot size represents the fraction of cells with expression.

The CITE-seq analysis combines single cell RNA-seq with simultaneous staining for a panel of >40 antibodies conjugated to uniquely barcoded oligonucleotides (henceforth Antibody derived tags, ADTs) aimed to cover the spectrum of normal hematopoietic stem (CD34⁺CD38^{−/low} fraction) and lineage progenitors (CD34⁺ enriched fraction, HSPCs) as well as surface markers reported to enrich for CML stem cells (Herrmann et al., 2014; Jaras et al., 2010; Kinstrie et al., 2020; Landberg et al., 2018; Sadovnik et al., 2016; Warfvinge et al., 2017). Upon sequencing, the scRNA-seq and ADT-seq libraries provide concurrent readout of gene expression and abundance of surface proteins respectively. This multiomic approach surpasses scRNA-seq in cellular subtype characterization and is not constrained by the conundrum of spectral overlap of fluorophores in FACS (Stoeckius et al., 2017), and availability of metal isotopes in mass cytometry as noted previously (Gullaksen et al., 2019). Our sample set of nine CML patients are from various clinical studies with attendant information on BCR::ABL1 transcript level as per International Scale (IS %) available after 12 months of TKI therapy. These were retrospectively classified as optimal, treatment failures, and warning cases in accordance with the guidelines from European LeukemiaNet (ELN) (Hochhaus et al., 2020) (Supp. Table 1).

To build a normal reference, we used our recently described CITE-seq profiles of Lin[−]CD34⁺ cells from normal BM from two age-matched healthy donors (henceforth, nBM) (Sommarin et al., 2021). The normal reference of 4696 Lin[−]CD34⁺ cells was visualized in a Uniform Manifold Approximation and Projection (UMAP, Fig. 1B). Although single cell omics datasets generate expression values for several thousand genes (dimensions), not all are equally informative. UMAP is a non-linear algorithm for dimensionality reduction and preserves the global data structure and enables visualization of the dataset in two dimensions. The nBM cells were subsequently clustered and annotated based on marker genes as well as information from antibody derived tags (ADTs) against surface markers (Fig. 1C–D, Supp. Table 4). We identified eleven cell clusters consisting of primitive cells, lineage progenitors and cycling cells in the UMAP, with near to equal distribution across the two nBM samples (Supp. Fig. 1). The primitive cluster was marked by expression of stem-like gene expression e.g., CRHBP, HLF as well as a CD90⁺CD38^{−/low} surface profile. Closest to the primitive cluster were MPP1 and 2 without frank manifestation of lineage

specific markers. The MPP1 cluster, in turn, was connected to the clusters with gain of myeloid-like, and lymphoid-like gene expression programs. This included myelo/lymphoid (My/Ly), myeloid (My: MPO, CD33), and lymphoid/dendritic/monocyte (Ly/pDc/mono) clusters (Ly86, CD135). The MPP2 cluster, in comparison, was connected to the megakaryocyte and erythroid lineage progenitors such as megakaryocyte-erythroid (MEP), megakaryocyte (MkP: ITGA2B, CD41), erythroid progenitors (ErP: HBA1, CD71, CD105) and basophil/mast cell clusters (Baso/mc: HDC, MS4A2-3, ITGB7).

Following CITE-seq of 14,274 Lin⁻CD34⁺ cells from nine CML patients, the cells were projected onto the nBM reference and annotated by label-transfer. Subsequently, we generated UMAPs for each patient and color-coded cells according to the transferred labels enabling us to visualize and highlight cellular heterogeneity on an individual basis (**Fig. 2A** [↗](#), **Supp. Fig. 2A** [↗](#)). A comparison of abundance of clusters showed that CML patients, in general, were enriched for basophil/mast cell, MEP, MkP clusters and depleted for primitive, MPP1, 2, and ly/pDC/Mono clusters (**Fig. 2B** [↗](#)). Inspection of profiles per patient showed that while most of the clusters were consistently enriched or depleted, ErP progenitors and cycling cells showed highly individual patterns (**Supp. Fig. 2B** [↗](#)). Furthermore, visual inspection of the individual UMAPs revealed an intriguing and distinct separation of cells within the primitive cluster in most of the patients (**Fig. 2A** [↗](#)). Taken together, CITE-seq and label transfer through nBM reference cell projection could successfully define the molecular heterogeneity within the Lin⁻CD34⁺ LSPC at diagnosis and revealed an average increase in abundance of MEP, MkP, and basophil/mast cells as compared to nBM.

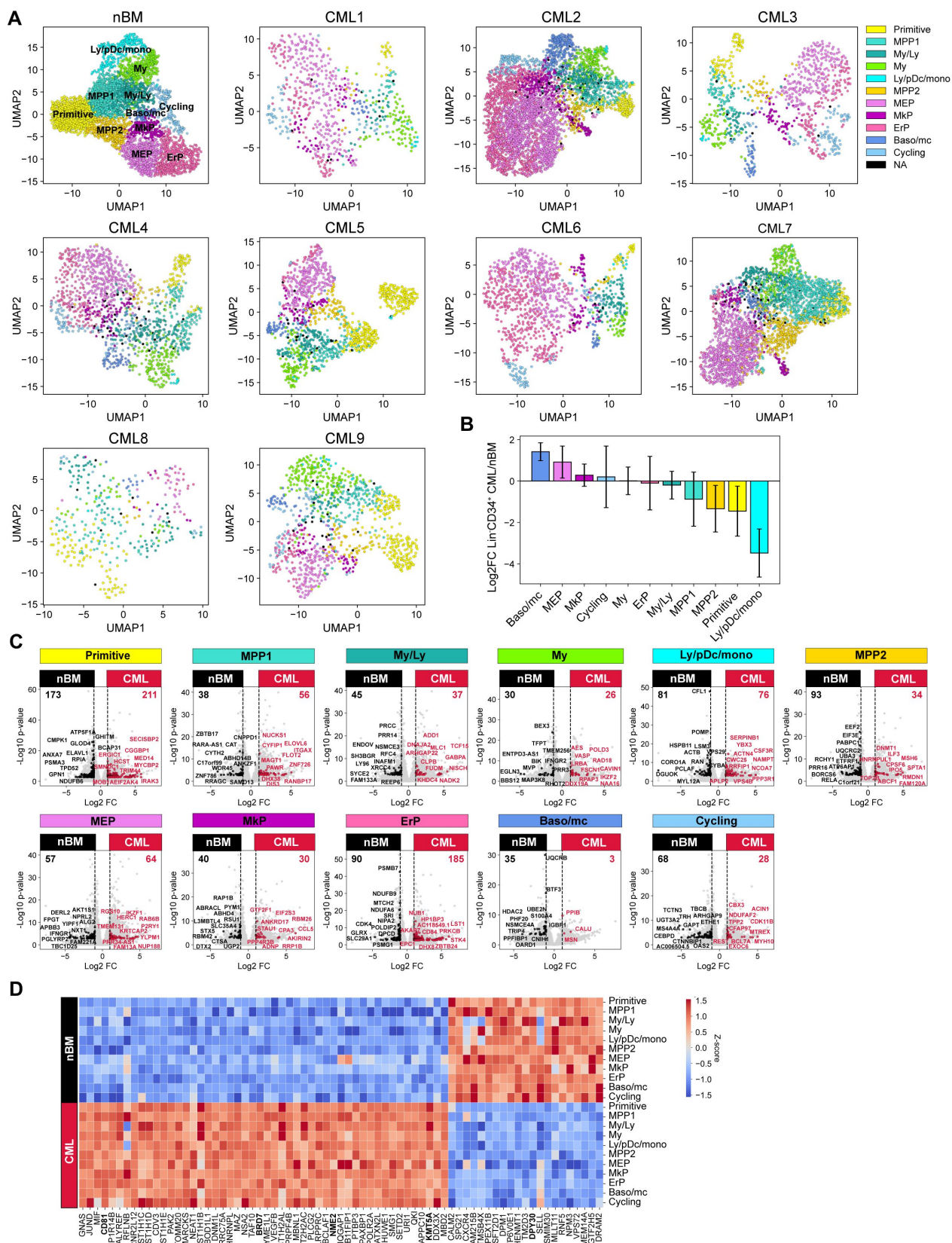


Figure 2.

Single cell maps of heterogeneity across patients and identification of a pan-stem and progenitor gene signature for CML

(A) UMAP embedding of Lin[−]CD34⁺ sorted CML BM cells from nine CML patients at diagnosis (14, 274 cells across patients). Cell color indicates cluster identity after label transfer through cell projection onto an aged-matched nBM reference (the first UMAP from the left).

(B) Bar plots showing the log2 fold change in cluster distribution (%) of Lin[−]CD34⁺ CML BM cells (n = 9) compared to Lin[−]CD34⁺ from nBM. Error bars depict standard deviation.

(C) Volcano plots showing differentially expressed genes between the Lin[−]CD34⁺ CML clusters (across all patients, n = 9) and the corresponding clusters from nBM (adjusted p-value < .01, log2 fold change > 1/< −1). Red and black dots represent genes uniquely up- or down-regulated per cluster comparison (genes not found significantly changed in any other cluster DEG analysis). The top 10 significant, unique genes are labelled in plot. Vertical dotted lines mark a log2 fold change equal to 1 and −1.

(D) Heatmap showing the average expression of the 50 significantly up-regulated and 21 down-regulated CML signature genes (genes significantly changed and consistent through all clusters DEG analyses) across clusters from all CML patients as well as nBM (adjusted p-value < .01, log2 fold change > 1/< −1). Red indicates high expression, blue low expression.

Identification of gene signatures unique to CML cells and shared across CML stem and progenitor populations

The identification of nBM and CML primitive populations and lineage progenitors prompted us to systematically compare pair-wise gene expression differences between all eleven clusters in CML and their nBM counterparts (**Fig. 2C**, Supp. Table 5). For each pair-wise comparison, we highlighted only the genes that were differentially expressed between the pair but not shared with any other cluster thereby defining strict cluster-specific changes. Although this revealed unique changes across all cluster pairs, the most striking changes were seen between primitive clusters from CML versus nBM with 384 differentially expressed genes. Moreover, erythroid progenitors also displayed large changes followed by ly/pDC/mono and ErP progenitors.

In contrast, when considering genes differentially expressed in CML relative to nBM, and crucially shared among all the cluster comparisons, we identified a set of 71 genes that comprised a pan-cml signature. This approach is preferable to a ‘bulk’ comparison of all CML CD34⁺ cells lumped together versus nBM cells, where the most numerically abundant cluster, and the most highly expressed genes therein, are likely to contribute disproportionately (**Fig. 2D**). Of the 71 genes, 50 genes were found to be consistently up-regulated in CML stem and progenitors. Notably, the individual genes from this set were expressed at varying levels in individual clusters (**Supp. Fig. 3A**, Supp. Table 6), reflecting a heterogenous origin of the CML signature. This included a constellation of genes such as surface markers, CD81 (Quagliano et al., 2020), nucleoside diphosphate kinase with transcriptional regulatory activity, NME2 (Kar et al., 2012; Thakur et al., 2009; Thakur et al., 2014; Thakur et al., 2011; Tschiedel et al., 2012; Tschiedel et al., 2008; Yadav et al., 2014) previously implicated in CML, genes of histone family, and components of SWI/SNF chromatin remodeling complex, BRD7 (Kaesler et al., 2008), as well as genes of histone methyltransferase family, KMT3A and 5A (Husmann & Gozani, 2019). The

analysis also identified 21 genes specifically and consistently upregulated in all clusters across nBM versus CML, and among others included DPY30 gene implicated in maintenance of adult HSCs (Yang et al., 2016 [DOI](#)).

Progenitor heterogeneity at diagnosis is linked to TKI therapy outcome

To assess how cellular heterogeneity within the Lin[−]CD34⁺ compartment at diagnosis is related to therapy response, the patients were stratified according to BCR::ABL1^{IS} (%) following 12 months of TKI therapy as per ELN recommendations (Supp. Table 1). The stem and progenitor heterogeneity at diagnosis was then evaluated relative to the treatment outcome by assessing cell composition of the individual CML patients 1-9 classified as optimal (CML1-4), warning (CML 5-7), and treatment failure (CML 8-9).

By enumerating the proportion of molecularly defined clusters of individual patients, we found that all clusters from the normal bone marrow reference samples were represented in each patient. However, there was remarkable heterogeneity between patients in terms of relative abundance of constituent clusters (**Fig. 3A** [DOI](#), **Supp. Fig. 2B** [DOI](#)). Importantly, this inter-patient heterogeneity followed patterns connected to therapy outcome. A focused comparison between optimal responders (CML 1-4) and treatment failure cases (CML 8-9) showed distinctive enrichment of specific cell clusters (**Fig. 3B** [DOI](#)), where failure cases displayed a higher abundance of the molecularly defined primitive cluster, multipotent progenitors 1 and 2, and myelo-lymphoid (My/Ly) and Ly/pDC/monocyte clusters at diagnosis. In contrast, the optimal responders had a higher burden of megakaryocyte-erythroid progenitors (MEP), megakaryocyte progenitors (MkP), Erythroid progenitor (ErP), Baso/mc and cycling clusters at diagnosis. Among others, the most conspicuous changes between optimal responders and treatment failures were observed in the primitive cluster (~4 fold higher in failures), and Meg-erythroid progenitors (MEP) and erythroid cluster (~3 fold higher in optimal). The warning cases (CML 5-7) also showed heterogeneity in terms of cell composition (**Fig. 3A** [DOI](#)); for instance, while CML 5 had a relatively higher content of primitive cells akin to treatment failures, CML 6 and 7 had a profile resembling optimal responders. Thus, these observations indicate that the prospective treatment outcome is reflected by the composition of stem and progenitor cell types at diagnosis.

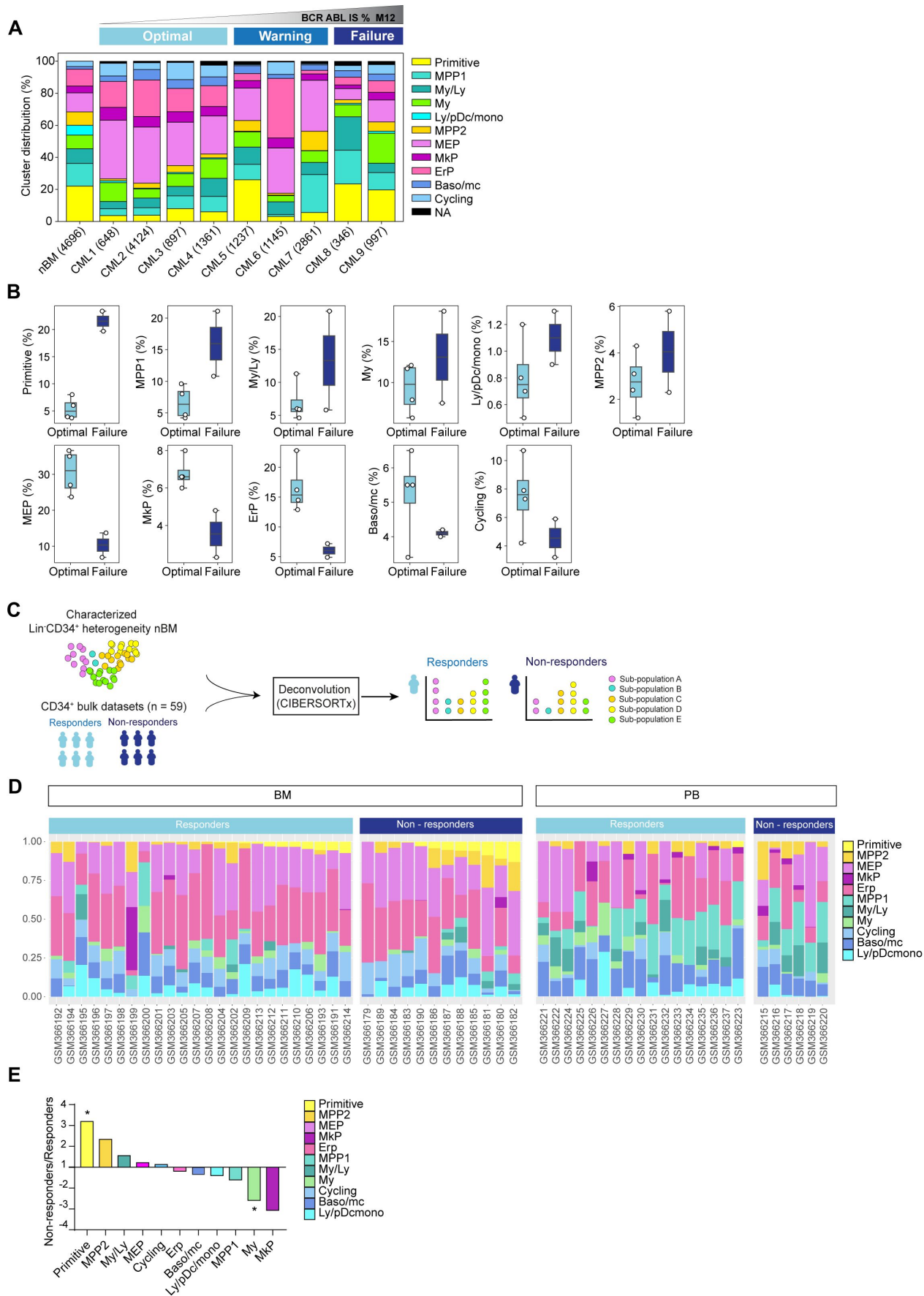


Figure 3.

Cell heterogeneity at diagnosis relates to TKI therapy outcome in CML

(A) Stacked bar plots showing the cluster distribution within Lin[−]CD34⁺ compartment in all CML patients at diagnosis as well as nBM. CML patients are ordered by BCR::ABL1^{IS} (%) following 12 months of TKI therapy (M12) and retrospectively stratified as per European LeukemiaNet recommendations; Optimal = CML1-4 ($\leq 0.1\%$), Warning = CML5-7 ($> 0.1-1\%$), Failure = CML8-9 ($> 1\%$). The total number of Lin[−]CD34⁺ cells from individual patients is indicated at the bottom.

(B) Box plots comparing cluster proportions in optimal responders (n = 4, CML patients 1-4) and treatment failures (n = 2, CML patients 8-9) at diagnosis; the patients were retrospectively stratified according to BCR::ABL1^{IS} (%) after 12 months of TKI therapy (Optimal $\leq 1\%$, Failure $> 1\%$).

(C) A scheme for computational deconvolution of bulk transcriptomes from patients into constituent cell populations; using CITE-seq derived gene signatures from Lin[−]CD34⁺ nBM as reference, an independent bulk CD34⁺ microarray dataset from CML patients (n = 59) (McWeeney et al., 2010) was deconvoluted into constituent cell populations using CIBERSORTx (Newman et al., 2019).

(D) Stacked bar plots showing percentage of specific clusters within CD34⁺ cells from individual CML patients (n = 59). The x-axis shows individual GSE ID for patients, y-axis shows percentage of clusters with similar color code as used in Fig. 1 UMAPs. Annotation of individual patient as per the original study (McWeeney et al., 2010); non-responders with $> 65\%$ Ph⁺ metaphases (did not achieve even a minor cytogenetic response), responders with 0% Ph⁺ metaphases after 12 months of Imatinib therapy (achieved CCyR). BM and PB represent CD34⁺ cells isolated from bone marrow, and peripheral blood respectively.

(E) The fold change in proportions for cell clusters between non-responders and responders (annotation as described above) for CD34⁺ cells isolated from bone marrow; statistical significance shown by asterisk *; student t-test, p-value $< .05$.

Large scale deconvolution of independent dataset shows cell composition is linked to cytogenetic response to TKI therapy

To further assess whether the hematopoietic stem and progenitor composition of CML patients at diagnosis is connected to their therapy response, we utilized an independent publicly available bulk gene expression dataset from CML patients with known therapy response (n=59) (McWeeney et al., 2010). This was done with an aim to computationally deconvolute the individual profiles into constituent cell types by employing the same nBM used to define the heterogeneity in the CITE-seq data as a molecular reference, subsequently estimate their relative abundances, and relate to TKI response (Fig 3C). We used CIBERSORTx (Newman et al., 2019), a widely used algorithm, which has also been recently used to deconvolute bulk transcriptomes from acute myeloid leukemia (AML) (Zeng et al., 2022). As the gene expression profiles were measured at diagnosis from CD34⁺ enriched cells from either bone marrow or peripheral blood, this represented a cellular fraction broadly similar to ours for comparison. Importantly, the CML patient cohort was stratified by the percentage of Ph⁺ cells after 12 months of Imatinib treatment as responders (0% Ph⁺ metaphases, complete cytogenetic response, CCyR) or non-responders ($> 65\%$ Ph⁺ metaphases, lack of even a minor cytogenetic response) as per the original study (McWeeney et al., 2010). We adhered to the patient annotation (responder or non-responder) provided by the original authors for consistency (see methods for more details).

The deconvolution results demonstrated a heterogeneous cell composition across patients, and overall comparison of imatinib non-responders ($n = 18$ out of 59) versus responders ($n = 41$ out of 59) showed a >3 fold statistically significant enrichment of primitive cells in non-responders (**Fig. 3D**, **Supp. Fig. 3B**). To assess whether the source of origin of $CD34^+$ cells in the public dataset could bias our results, we compared the abundances of imputed cell types from non-responders and responders based on their extraction from either bone marrow, or peripheral blood separately. This once again revealed a statistically enriched presence of primitive cells in non-responders versus the responders in the bone marrow (~ 3 fold higher). Moreover, and unlike the findings from overall comparison, myeloid cells were also found to be statistically enriched within responders (~ 2.5 fold higher) (**Fig. 3E**). Although MPP2 and MkP clusters showed increased abundance in non-responders and responders respectively, the difference did not reach statistical significance. Notably, the primitive cells were rarely detected in peripheral blood in contrast to the bone marrow, confirming the notion that cell type fractions differ in different tissues (**Fig. 3D**), and the bone marrow is predictably a more reliable and abundant source of true primitive cells in CML. Taken together, our analyses revealed distinct cellular heterogeneity across CML patients at diagnosis and highlighted that cell composition is associated with therapy response. Specifically, a higher burden of cells with a stem-like signature at diagnosis is a strong indicator of treatment failure.

Identification of $BCR::ABL1^+$ and $BCR::ABL1^-$ primitive cells and their surface markers by combining single-cell gene expression with ADTs

Interestingly, according to the ADT data the established CML stem cell markers CD25, CD26, and CD93 could only capture a fraction of the primitive cells (**Supp. Fig. 3C**). To further investigate the molecular and immunophenotypic heterogeneity within the primitive cluster we performed CITE-seq on the stem cell enriched $Lin^-CD34^+CD38^{-/low}$ compartment from the same patients ($n = 8$; 6,779 cells). A stepwise analysis (**Fig. 4A**) was performed to resolve the $BCR::ABL1$ status of the individual primitive cells and determine their immunophenotype. First (1), the primitive cells from the $Lin^-CD34^+CD38^{-/low}$ fraction of all CML patients were merged and classified by $BCR::ABL1$ status using publicly available $BCR::ABL1^+$ gene signatures. Next (2) CITE-seq data from both the Lin^-CD34^+ and the $Lin^-CD34^+CD38^{-/low}$ sorted population from the same patient was merged and visualized in a joint UMAP, while (3) the $BCR::ABL1$ status of the $Lin^-CD34^+CD38^{-/low}$ primitive cells was linked to the merged UMAPs by cell ID. Finally (4) the log2 fold change in ADT expression between $BCR::ABL1^+$ and $BCR::ABL1^-$ primitive cells were calculated and the remaining cells of the UMAP were color-coded by the cluster identity generated by label-transferred projection to nBM.

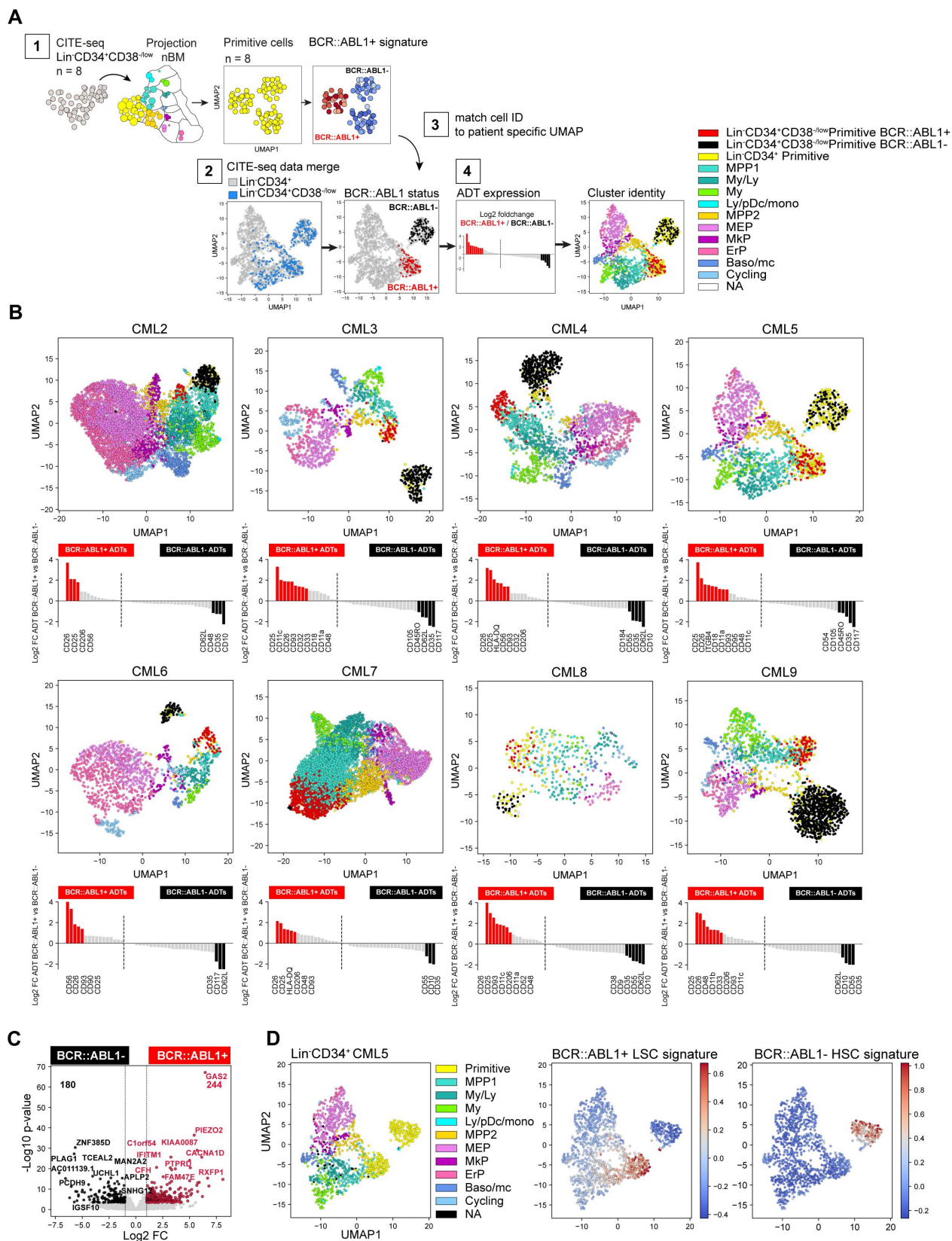


Figure 4.

Identification of BCR::ABL1⁺ and BCR::ABL1⁻ primitive cells and their surface markers by joint analysis of single cell gene expression and multiplexed antibody derived sequence tags (ADTs)

(A) Overview of analysis steps to generate **Figure 4B**. (1) Lin⁻CD34⁺CD38^{-/low} cells were projected onto Lin⁻CD34⁺ nBM. Subsequently, the primitive cluster CITE-seq data from all CML patients and nBM (negative control) were merged and visualized together in one UMAP. BCR::ABL1⁺ gene signatures from BCR::ABL1 targeted SMART-seq (Giustacchini et al., 2017) was used to define primitive cluster cells from the individual patients as either BCR::ABL1⁺ or BCR::ABL1⁻ (2) Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} CITE-seq data from the same patient was merged and visualized together in a UMAP (3) BCR-ABL status from the Lin⁻CD34⁺CD38^{-/low} primitive cells was linked to the joint UMAPs by matching cell IDs and cell were colored as either red (BCR::ABL1⁺) or black (BCR::ABL1⁻) (4) The log2 fold change in ADT expression between primitive Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁺ (red) and BCR::ABL1⁻ (black) cells was calculated and the remaining cells were colored according to their cluster annotation given after projection onto Lin⁻CD34⁺ nBM.

(B) UMAP plots showing the merged CITE-seq data of Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} sorted populations for CML2-9 individually. Cells are color-coded according to cluster annotation given following projection onto Lin⁻CD34⁺ nBM. Lin⁻CD34⁺CD38^{-/low} primitive cluster cells are annotated as BCR::ABL1⁺ (red) or BCR::ABL1⁻ (black) and display enrichment of BCR::ABL1⁺ LSC signatures and non-leukemic stem cell signatures respectively. The red and black bars in the bar plot below indicate ADTs with significant changes in expression (p-value < .05, log2 fold change >1 / <-1) for BCR::ABL1⁺ and BCR::ABL1⁻ cells, respectively.

(C) Volcano plot showing differentially expressed genes between CML Lin⁻CD34⁺CD38^{-/low} Primitive BCR::ABL1⁺ and BCR::ABL1⁻ cells. Red and black dots represent significant up- and down-regulated genes (adjusted p-value < .01, log2 fold change >1 / <-1). The top 10 significant genes are labelled. Vertical dotted lines mark a fold change equal to 1 and -1.

(D) UMAP embedding of CML Lin⁻CD34⁺ cells from one representative CML patient at diagnosis (CML5). First plot from the left show cells colored according to the cluster identity given after mapping to nBM, the primitive cells are colored yellow. The following two UMAPs of Lin⁻CD34⁺ show the relative mRNA expression of the BCR::ABL1⁺ LSC signature and BCR::ABL1⁻ non-leukemic stem cell gene signature; scale: red = high expression, blue = low expression.

The BCR::ABL1 status of the primitive cells was determined using two previously established gene expression signatures comparing Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁺ LSCs to normal Lin⁻CD34⁺CD38^{-/low} HSCs, and Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁻ non-leukemic stem cells (Giustacchini et al., 2017). These signatures have been derived by analyzing cells with SMART-seq for global transcriptome, and primers specific to the BCR::ABL1 fusion gene, simultaneously and therefore represent useful anchors to query and label cells as either BCR::ABL1⁺ or BCR::ABL1⁻ (See methods for detailed analysis).

In the UMAP generated from all Lin⁻CD34⁺CD38^{-/low} primitive cells, we identified three coarse clusters, where two consisted of primitive CML cells with representation from all patients (cluster 2-3), and one represented Lin⁻CD34⁺CD38^{-/low} primitive cells from nBM which were included as negative control (cluster 1) (**Supp. Fig. 4B-C**). When we queried each cluster for enrichment of either LSC or HSC-like gene signatures the analysis clearly demonstrated that the primitive cells from CML samples could be divided based on the presence of BCR::ABL1, where CML cluster 3 displayed an evident enrichment of BCR::ABL1⁺ LSC specific signatures as compared to CML cluster 2 consisting of BCR::ABL1⁻ non-leukemic stem cells (**Supp. Fig. 4D-E**). Interestingly, both CML clusters were also found to share gene characteristics with normal stem cells (**Supp. Fig. 4D**). Accordingly, we annotated Lin⁻CD34⁺CD38^{-/low} primitive cells from individual patients as

either BCR::ABL1⁺ or BCR::ABL1⁻ and could observe that the distribution varied across patients, suggesting a variable load of leukemic stem and normal stem-like cells within the primitive fraction (Supp. Fig. 4F [↗](#)).

Remarkably, subsequent matching of BCR::ABL1⁺ and BCR::ABL1⁻ cell IDs to the primitive cells in the joint Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} UMAPs (indicated by red and black cell color, respectively), indeed revealed that BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs were distinctly and consistently separated in the UMAP for each patient (Fig. 4B [↗](#), Supp. Fig. 5 [↗](#)). BCR::ABL1⁺ LSCs in all patients were positioned immediately juxtaposing the downstream progenitor populations forming a continuum of differentiation hierarchy characteristically observed in single-cell analysis of HSPCs (Pellin et al., 2019 [↗](#); Safi et al., 2022 [↗](#); Velten et al., 2017 [↗](#)). In contrast, BCR::ABL1⁻ HSCs formed an isolated and distinct cluster clearly detached from the rest of the cells in the majority of patients, indicating that during the chronic phase of CML, active hematopoiesis is dominated by the BCR::ABL1⁺ LSCs while BCR::ABL1⁻ HSCs reside in the bone marrow albeit in an inactive state with reduced contribution to hematopoiesis as previously suggested (Chen et al., 2023 [↗](#); Coulombel et al., 1983 [↗](#)). Moreover, using the ADT data we could validate the molecularly defined BCR::ABL1⁺ LSCs also by surface expression, with CD26, CD25, and CD93 displaying a 2-8 fold elevated expression in BCR::ABL1⁺ primitive cells as compared to BCR::ABL1⁻. A noteworthy patient-specific variability in surface protein expression on the most primitive cells was also established (Fig. 4B [↗](#)).

To gain understanding of the nature of molecular program of BCR::ABL1⁺ vs. BCR::ABL1⁻ primitive cells, we analyzed differentially expressed genes between the pooled primitive BCR::ABL1⁺ and BCR::ABL1⁻ populations from all CML patients. This identified 244 genes specifically upregulated in BCR::ABL1⁺ and 180 genes upregulated in BCR::ABL1⁻ cells (Fig. 4C [↗](#), Supp. Table 7). Interestingly, among genes up-regulated in BCR::ABL1⁺ primitive cells included the expression of established CML markers; IL2RA (CD25), DPP4 (CD26) and IL1RAP as well as LEPR (CD295) a receptor recently described as a novel marker for Lin⁻CD34⁺CD38^{-/low} LSCs (Landberg et al., 2018 [↗](#)). In contrast, BCR::ABL1⁻ cells displayed an up-regulated expression of the stem cell related gene CRHBP (Barbieri et al., 2022 [↗](#); He et al., 2005 [↗](#)) as well as CXCR4 important for homing and maintenance of normal HSCs (Sugiyama et al., 2006 [↗](#)). Upon applying the gene signatures to Lin⁻CD34⁺ cells, we could clearly identify as well as partition primitive cells into BCR::ABL1⁺ vs. BCR::ABL1⁻ (Fig. 4D [↗](#) and Supp. Fig. 6 [↗](#)). This is especially relevant as the Lin⁻CD34⁺ fraction contains a higher proportion of lineage progenitors but a smaller fraction of primitive cells making their identification challenging. Taken together, this multistep analysis approach generated unique detailed UMAPs allowing for inspection of BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs, their relation to other progenitor cells, as well as further validate BCR::ABL1⁺ LSCs identity by surface protein expression.

A unique combination of CD26 and CD35 surface markers captures molecularly defined BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs

A long-sought aim in CML research has been to generate protocols for effective discrimination and isolation of CML LSCs from residual HSCs within individual patient bone marrow. The single cell multiomic feature space of CITE-seq together with the identification of molecular distinct BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs provides a unique opportunity to identify efficient protocols for their separation.

To specifically define the surface markers for their ability to capture BCR::ABL1⁺ vs. BCR::ABL1⁻ primitive cells, we compared significantly up-regulated ADTs for either Lin⁻CD34⁺CD38^{-/low} BCR::ABL1⁺ LSCs or BCR::ABL1⁻ HSCs cells across patients and identified markers with consistent expression across the data set. We found 28 unique markers to be significantly up-regulated for either BCR::ABL1⁺ or BCR::ABL1⁻ cells (black or red colored bars in Fig. 4B [↗](#), Fig. 5A [↗](#)). Of these 28 markers, 17 ADTs marked BCR::ABL1⁺ LSCs and were detected in at least one patient. However,

only CD26, and CD25 were explicit and consistent throughout the entire cohort. CD93 was present on BCR::ABL1⁺ LSCs in 7/8 patients and other previously suggested LSC markers like CD33 and CD56 specifically labeled BCR::ABL1⁺ primitive cells in only a fraction of the patients. The documented CML LSC marker IL1RAP was not in the ADT panel and hence could not be probed. In contrast, 11 ADTs were specifically detected in BCR::ABL1⁻ primitive cells from at least one patient. Interestingly, CD35 with ~4-fold elevated expression as compared to BCR::ABL1⁺ LSCs was the only marker found to be consistently specific for the BCR::ABL1⁻ HSC population across all patients. CD62L and CD10 were specifically present on BCR::ABL1⁻ HSCs in 6/8 and 5/8 patients, respectively. Intriguingly, we recently showed that CD35 captures all HSCs in healthy human hematopoiesis and that stem cell activity as measured by xenotransplantation assays and epigenetic profiling is restricted to CD35⁺ HSCs (Sommarin et al., 2021 [DOI](#)). By applying the BCR::ABL1 signatures also to the Lin⁻CD34⁺ primitive cells (**Supp. Fig. 7** [DOI](#)), we could visualize the relative expression within the Lin⁻CD34⁺ UMAPs in relation to the cells BCR::ABL1 status and further confirm that the primitive fraction, previously captured by the established CML stem cell markers CD25, CD26, and CD93, consisted of specifically BCR::ABL1⁺ LSCs (**Fig. 5 B-C** [DOI](#), **Supp. Fig. 8** [DOI](#) and **9A** [DOI](#)). In addition, we observed that our previously described surface marker CD117, whose absence characterized LSCs persisting 3 months of TKI therapy, displayed a highly variably expression at diagnosis both marking a proportion of LSCs and HSCs across patients. Interestingly, despite that relative expression of CD35⁺ was lower in the primitive cells compared to MEP/ERP cell clusters the CD35⁺ gate captured ~50% of the BCR::ABL1⁻ primitive cluster.

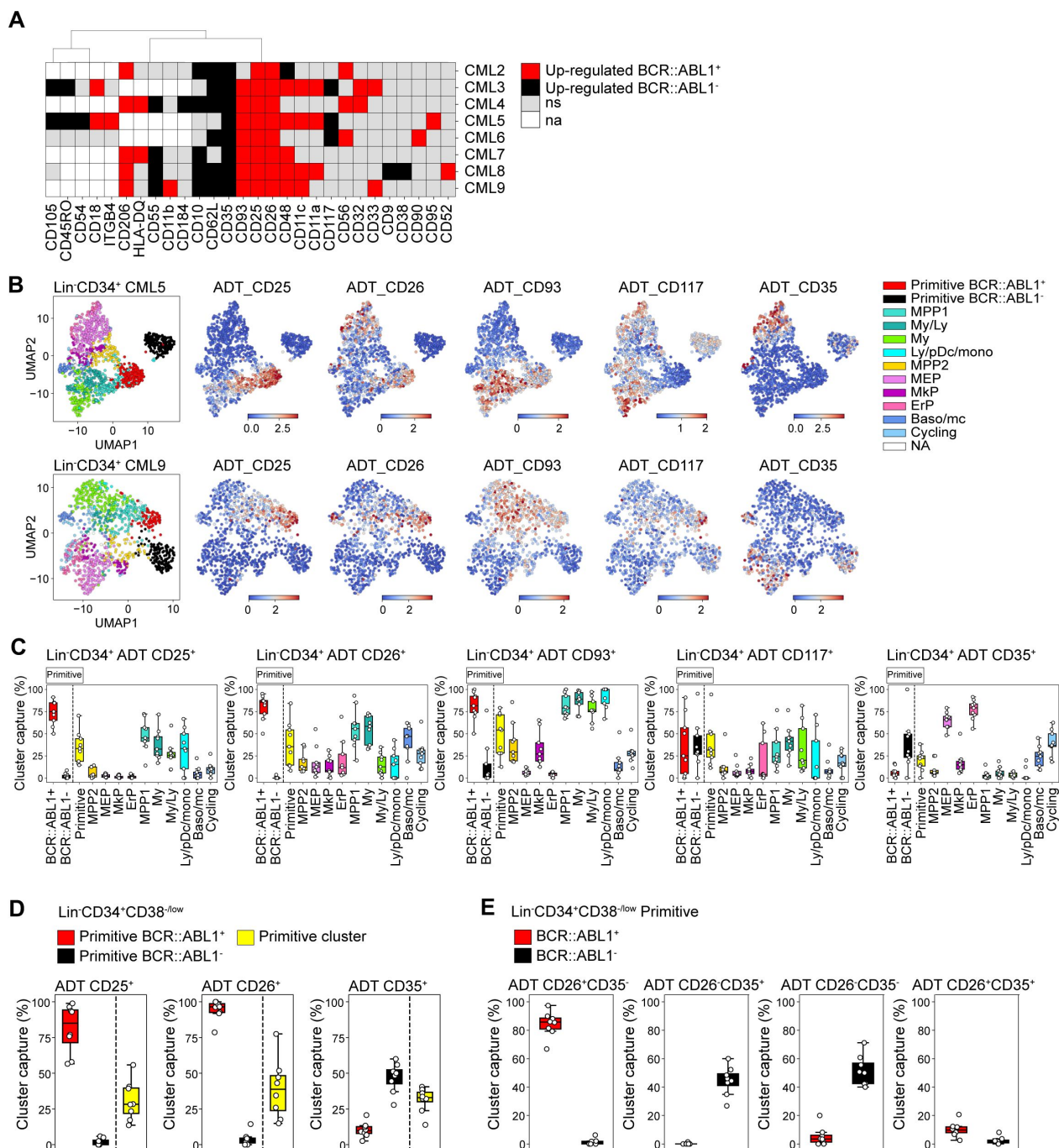


Figure 5.

Assessment of surface marker combinations to capture molecularly defined BCR::ABL1⁺ and BCR::ABL1⁻ primitive cells

(A) Heatmap comparing ADT expression between BCR::ABL1⁺ and BCR::ABL1⁻ cells within the Lin⁻CD34⁺CD38^{-/low} primitive cluster across CML patients (black = surface markers significantly up-regulated in BCR::ABL1⁻ cells (log2 fold change < -1, p-value < .05); red = surface markers significantly up-regulated in BCR::ABL1⁺ cells (log2 fold change > 1, p-value < .05); grey = non-significant change in ADT expression; white = surface marker was not present in the ADT panel used for the specific patient).

(B) Visualization of the relative ADT expression of CD25, CD26, CD93, CD117 and CD35 in the UMAPs of Lin⁻CD34⁺ cells from CML patient 5 and 9 (red = high expression, blue = low expression).

(C) Assessment of a selection of ADTs; CD25, CD26, CD93, CD117 and CD35 capacity to capture BCR::ABL1⁺ primitive, BCR::ABL1⁻ primitive, all primitive cells, and specific progenitors across patients by gating on their expression within the Lin⁻CD34⁺ compartment (x-axis shows specific clusters, y-axis: percentage of specific cluster captured; each white circle represents a patient sample).

(D) Assessment of specific ADTs: CD26, CD25 and CD35 capacity to capture BCR::ABL1⁺ primitive versus BCR::ABL1⁻ primitive cluster within the Lin⁻CD34⁺CD38^{-/low} compartment across patients; x-axis shows specific clusters, y-axis: percentage of specific cluster captured; each white circle represents a patient sample.

(E) Assessment of specific combinations of ADTs: CD26⁺CD35⁻, CD26⁻CD35⁺, CD26⁻CD35⁻, and CD26⁺CD35⁺ capacity to capture BCR::ABL1⁺ primitive versus BCR::ABL1⁻ primitive cluster within the Lin⁻CD34⁺CD38^{-/low} compartment across patients; x-axis shows specific clusters, y-axis: percentage of specific cluster captured; each white circle represents a patient sample.

Given the consistency of detection across patients, we specifically focused on the ability of ADTs against CD25, CD26 and CD35 to separate between BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs within the stem cell-enriched Lin⁻CD34⁺CD38^{-/low} compartment. CD26 showed the highest capture of the BCR::ABL1⁺ primitive cluster (>95%) with a consistent capture percentage across patients (**Fig. 5D**). This was followed by CD25 which captured >80% of the BCR::ABL1⁺ primitive cluster on average but displayed higher variability in capture efficiency compared to CD26 across patients. In comparison, the percentage of BCR::ABL1⁻ primitive cluster captured by CD26 and CD25 was considerably lower (~5%). In contrast, the CD35⁺ gate showed a remarkable specific capture of BCR::ABL1⁻ primitive cluster (~50%) (**Fig. 5D**).

To further confirm whether CD26 and CD35 together could efficiently separate BCR::ABL1⁺ and BCR::ABL1⁻ primitive cells, we gated for different combinations of CD26 and CD35 within the Lin⁻CD34⁺CD38^{-/low} compartment (**Fig. 5E**, **Supp. Fig. 9B**). Predictably, whereas a combination of CD26⁺CD35⁻ ADTs captured BCR::ABL1⁺ primitive LSC cluster at high purity, the CD26⁻CD35⁺ combination marked BCR::ABL1⁻ primitive HSC cluster with no contamination of BCR::ABL1⁺ LSCs (**Fig. 5E**). Interestingly, CD35 expression divided the CD26⁻, BCR::ABL1⁻ primitive cells in one CD35⁺ and one CD35⁻ fraction. This is in line with our recent observation that human BM immunophenotypic stem cells can be divided by CD35, where all HSC activity is captured within the CD35⁺ fraction (Sommarin et al., 2021). Accordingly, ADT combination of CD26⁻CD35⁻ captured BCR::ABL1⁻ primitive cells likely representing non-leukemic MPPs downstream of normal CD26⁺CD35⁺ stem cells. Notably, only a minority of cells were positive for ADTs for the CD26⁺CD35⁺ combination as evidenced by an overall low capture of cells in general. Thus, this shows the importance for positive, HSC-specific markers for efficient isolation of CML LSCs, and detection of residual HSCs in leukemia. Taken together, these observations suggested

that although patients are heterogeneous for a given marker, BCR::ABL1⁺ LSCs can be consistently captured by the CD26⁺CD35⁻ combination while BCR::ABL1⁻ HSCs expressed a CD26⁻CD35⁺ immunophenotype within Lin⁻ CD34⁺CD38^{-/low} compartment in the bone marrow of CML patients.

A comparative analysis of CD26⁺CD35⁻ versus CD26⁻CD35⁺ cells for BCR::ABL1 expression and response to TKI therapy

To explore whether the combination of CD26 and CD35 could indeed purify BCR::ABL1⁺ vs. BCR::ABL1⁻ in Flow Cytometry protocols, we sorted CD26⁺CD35⁻ and CD26⁻CD35⁺ cells from the Lin⁻CD34⁺CD38^{-/low} compartment from patients both before and following 3 months of Bosutinib therapy and employed quantitative real time qPCR analysis using Taqman probes against the BCR::ABL1 fusion gene (**Fig. 6A** [↗](#)). The analysis showed a strong signal within CD26⁺CD35⁻ cells for BCR::ABL1 expression while CD26⁻CD35⁺ cells were BCR::ABL1^{low/negative}, validating the capacity of the combination of CD26 and CD35 to separate BCR::ABL1⁺ and BCR::ABL1⁻ stem cells at diagnosis as well as after 2nd generation TKI therapy. In addition, real time qPCR analysis confirmed that Lin⁻CD34⁺CD38^{-/low}CD26⁺CD35⁻ cells from CML patients at diagnosis displayed a strong signal for GAS2 (**Fig. 6B** [↗](#)), one of the top genes from the CITE-seq BCR::ABL1⁺ LSC signature (**Fig. 4C** [↗](#)). Furthermore, CD26⁺CD35⁻ cells maintained GAS2 expression following 3 months of Bosutinib therapy (**Fig. 6B** [↗](#)). GAS2 has previously been linked to CML disease progression, cell cycle control, apoptosis, and response to Imatinib (Janssen et al., 2005 [↗](#); Radich et al., 2006 [↗](#); Zhou et al., 2014 [↗](#)).

To assess the level of quiescence between the BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs (Nakamura-Ishizu et al., 2014), we sorted Lin⁻CD34⁺CD38^{-/low}CD45RA⁻CD26⁺CD35⁻ and CD26⁻CD35⁺ single cells from two CML patients at diagnosis into individual wells and examined their division kinetics for 140 hours. However, LSC and HSC populations were indistinguishably deeply quiescent with average times to first division of 80-100 hour in culture (**Supp. Fig. 9C-D** [↗](#)). Moreover, cell cycle analysis was performed by gating the corresponding cell populations in the Lin⁻CD34⁺CD38^{-/low} CITE-seq data using ADTs. In accordance with the *in vitro* study, there was no distinct difference in cell cycle status between the populations (**Supp. Fig. 9E-F** [↗](#)).

The higher burden of primitive cells in treatment failures observed above prompted us to query to which extent patients differed in the load of isolatable BCR::ABL1⁺ vs. BCR::ABL1⁻ at diagnosis and their sustenance in response to TKI therapy. Intriguingly, we noted a substantially higher ratio of CD26⁺CD35⁻ LSCs over CD26⁻CD35⁺ HSCs at diagnosis in prospective treatment failures compared to the bone marrow of optimal responders, both using CITE-seq ADTs as well as in FACS analysis for CD26 and CD35 (**Fig. 6C-D** [↗](#), **Supp. Fig. 9B** [↗](#)). Importantly, following 3 months of TKI therapy, the percentage of BCR::ABL1⁺ CD26⁺CD35⁻ cells decreased in both groups of patients; however, there was a striking difference in the relative proportion of CD26⁺CD35⁻ LSCs/CD26⁻CD35⁺ HSCs in both groups. The stem cell compartment of optimal responders was dominated by healthy HSCs at three months following treatment. In contrast, the stem cell population of failure patients still comprised of CD26⁺CD35⁻ LSCs without re-establishment of HSCs and restoration of normal hematopoiesis (**Fig. 6E** [↗](#)). Taken together, we here for the first time present a protocol for separation and isolation of both BCR::ABL1⁺ LSCs and BCR::ABL1⁻ HSCs from the same CML patients, two populations evidently relevant to therapy response.

Discussion

As new and more potent TKIs continue to be developed (Braun et al., 2020 [↗](#)), the persistence of fully leukemogenic cells even in TFR necessitates life-long therapy in most patients. The ensuing treatment-emergent adverse events, and financial toxicity (Cortes et al., 2021 [↗](#); Lipton et al.,

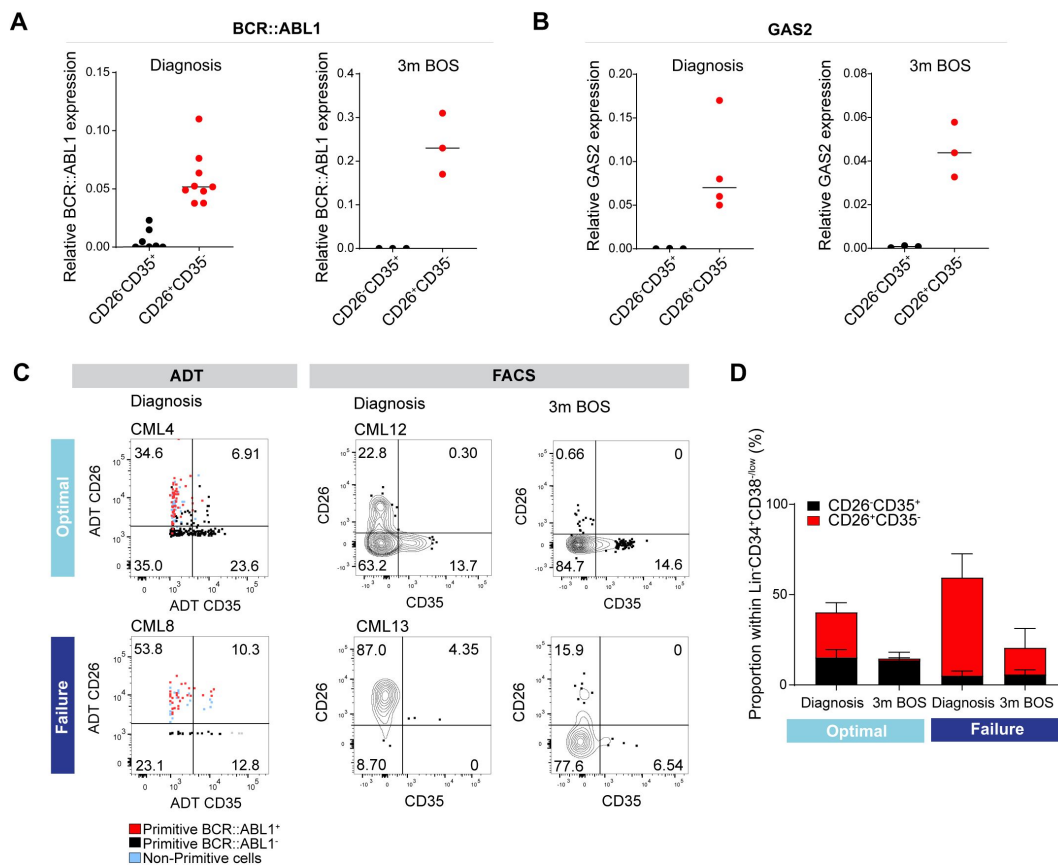


Figure 6.

Analysis of CD26⁺CD35⁻ vs. CD26⁻CD35⁺ cells for BCR::ABL1 transcript expression, and response to TKI therapy

(A) Real time qPCR for BCR::ABL1 in CD26⁻CD35⁺ (n = 7, CML patients 10-12, 14-15, 17-18) vs CD26⁺CD35⁻ (n = 9, CML patients 10-15, 17-18 and 20) within the Lin⁻CD34⁺CD38^{low} compartment at diagnosis and following 3 months of Bosutinib therapy (n = 3, CML patients 17, 18, 21). GAPDH served as control.

(B) Real time qPCR for GAS2 in CD26⁻CD35⁺ (n = 3, CML patients 14, 17-18) vs CD26⁺CD35⁻ (n = 4, CML patients 13-14, 17-18) within the Lin⁻CD34⁺CD38^{low} compartment at diagnosis and following 3 months of Bosutinib therapy (n = 3, CML patients 17-18, 21). GAPDH served as control.

(C) Representative FACS plots showing the percentage CD26⁺CD35⁻ (LSC) and CD26⁻CD35⁺ (HSC) cells within the Lin⁻CD34⁺CD38^{low} compartment. Left panel: ADT gated CD26⁺CD35⁻ and CD26⁻CD35⁺ cells within Lin⁻CD34⁺CD38^{low} CITE-seq data from selected optimal responder (CML4) and treatment failure (CML8) at diagnosis. Right panel: FACS gated Lin⁻CD34⁺CD38^{low}CD26⁺CD35⁻ and CD26⁻CD35⁺ cells in selected optimal responder (CML12) and treatment failure (CML13) at diagnosis versus 3 months of TKI therapy (Bosutinib).

(D) Bar plots showing the percentage of CD26⁺CD35⁻ and CD26⁻CD35⁺ cells within Lin⁻CD34⁺CD38^{low} compartment from optimal responders and treatment failures at diagnosis (n = 11; optimal = 6 (≤ 0.1 %, CML patients 12, 14-18), failure = 5 (> 1 %, CML patients 13, 19, 22-23, 25) and following 3 months of Bosutinib therapy (n = 11; optimal = 6 (≤ 0.1 %, CML patients 12, 14-18), failure = 5 (> 1 %, CML patients 13, 19, 22-23, 24) determined using FACS.

2022 [Zafar, 2016](#)) motivate a search for organizing principles to predict therapy response, dampen disease progression and achieve durable TFR in a larger fraction of CML patients. One recent development has been to invoke the presence of somatic mutations other than BCR::ABL1 to explain primary resistance to TKIs, and disease progression (Branford et al., 2019). A complementary perspective is to consider leukemic cell state and heterogeneity as a fundamental determinant of response to TKI to augment the mutational foundation, as has been recently demonstrated in AML (Zeng et al., 2022).

Our single cell multiomics maps show clear differences in overall cell composition within stem and progenitor compartments in leukemia patients at diagnosis versus nBM. Importantly, treatment failures and optimal responders displayed distinctive enrichment of specific cell clusters. Using our 11-molecularly defined clusters as anchors, we deconvoluted the bulk gene expression profiles from n=59 CML patients to infer constituent cell populations and found that a more profound stemness/primitive signature in the BM consistently associated with inferior therapy response. Indeed, there is growing recognition of the burden of primitive cells, and the ratio of BCR::ABL1⁺ vs. BCR::ABL1⁻ within the primitive compartment as potentially clinically relevant features such as hemoglobin, blast count, overall survival, progression free survival, and therapy response (Fathy El-Metwaly et al., 2021; Krishnan et al., 2023; Mustjoki et al., 2013; Thielen et al., 2016). Apart from the differential burden, the primitive cells from optimal and sub-optimal responders could also be qualitatively different e.g., in terms of gene expression profile as suggested recently (Krishnan et al., 2023). The relevance of BCR::ABL1⁺ LSC burden to 1st and 2nd generation TKI therapy outcome stands in contrast to the leukemic progenitors (Ph⁺ CD34⁺CD38⁺) which did not show such correlation (Mustjoki et al., 2013). Moreover, longstanding observations have revealed that CML patients at diagnosis contain both BCR::ABL1⁺ LSCs vs. BCR::ABL1⁻ HSC within the Lin⁻CD34⁺CD38^{-/low} immunophenotypic compartment, and BCR::ABL1⁺ LSCs suppress residual normal HSCs (Chen et al., 2023; Coulombel et al., 1983). Predictably, a lower fraction of BCR::ABL1⁻ normal stem cells at diagnosis and during therapy relates to hematological toxicity with delayed or compromised restoration of Ph^{neg} hematopoiesis upon successful response to TKIs (Janssen et al., 2012). Here, we for the first time present an effective means to separate CML LSCs from HSCs within the stem cell compartment of patients. Prospective optimal responders to TKI treatment had a higher content of BCR::ABL1⁻CD26⁻CD35⁺ cells at diagnosis, and 3 months following therapy versus the treatment failures. We have recently demonstrated that Lin⁻CD34⁺CD38^{-/low}CD45RA⁻CD35⁺ cells are at the top of human hematopoietic hierarchy and display chromatin accessibility and transcriptional enhancers in line with their multilineage long-term reconstitution ability (Sommarin et al., 2021).

With the high-throughput and resolution now provided by single cell methods, BCR::ABL1⁺ cells can be distinguished from BCR::ABL1⁻ cells residing within the same immunophenotypic compartment as described above, and importantly, heterogeneity within BCR::ABL1⁺ LSC subpopulations in terms of cell surface markers, molecular signature, and TKI response can be measured. The use of CITE-seq enabled generation of patient-specific maps providing a panoramic yet exquisitely detailed view of cellular heterogeneity. Intriguingly, one of the emerging principles from the single cell studies has been that not all LSC subpopulations are equally sensitive to BCR::ABL1 inhibition; while a majority of the LSCs are depleted, at least a fraction survives (Giustacchini et al., 2017; Warfvinge et al., 2017). This is corroborated by overall reduction in BCR::ABL1⁺ Lin⁻CD34⁺CD38^{-/low} cells but a striking enrichment of cKIT⁺CD26⁺ subpopulation displaying primitive and quiescent molecular program upon commencing TKI therapy (Warfvinge et al., 2017). Using CITE-seq, we confirmed the existence of these cells already at diagnosis and showed that BCR::ABL1⁺ vs. BCR::ABL1⁻ can be efficiently discriminated as CD26⁺CD35⁻ and CD26⁻CD35⁺ respectively within the Lin⁻CD34⁺CD38^{-/low} fraction, and thus facilitating improved assessment of LSC burden.

The inspection of single cell maps raises questions both regarding the origin of the interpatient heterogeneity in LSC burden as well as how these cells differentially manage to survive TKI therapy. Given the heterogeneity within normal HSCs with respect to lineage bias and clonal output (Haas et al., 2018 [DOI](#)), it is tempting to speculate that stem/progenitor cell acquiring the BCR::ABL1 oncogenic hit perhaps might be different across CML patients ensuring disparity in LSC load and characteristics. An equally plausible alternative is variability in the bone marrow niche that might favor differential abundance of stem/lineage-biased progenitors in patients (Baryawno et al., 2017 [DOI](#)). Once established, at least a subset of the LSCs is likely to be independent of BCR::ABL1 kinase signaling for their survival (Bhatia et al., 2003 [DOI](#); Corbin et al., 2011 [DOI](#); Hamilton et al., 2012 [DOI](#)). However, whether their self-sustenance involves switching to non-kinase activity of BCR::ABL1 protein, or achieving total BCR::ABL1 independence, and whether such mechanisms are inherently active in treatment naïve cells or become active only after exposure to TKIs is poorly understood (Zhao & Deininger, 2020 [DOI](#)).

We foresee future investigations to provide several key insights. First, accurate and direct detection of fusion transcripts such as BCR::ABL1 in single cells as part of scRNA-seq remains a major constraint, however, new approaches should allow direct measurement of BCR::ABL1 on a massive scale. Second, CITE-seq, a single cell multiome approach, like any other high-throughput single cell method comes with methodological, technical, and biological challenges. Cell sample handling, library preparation, sequencing quality and depth, drop-out of lowly expressed genes, and taking a molecular ‘snapshot’ of cells in time are all merely some factors that could introduce variability and noise to the data. A simpler, faster, and cheaper FACS panel detecting CD26⁺ and CD35⁺ surface markers can possibly work as an applicable clinical tool to quantify leukemic and non-leukemic stem cells in CML patients. Future prospective studies of this FACS panel coupled to clinical trials in large patient cohorts will establish the diagnostic and prognostic value of the relative abundance of these populations. It will be of importance to evaluate if CD26⁺CD35⁺ cells are critical for restoring normal hematopoiesis once the TKI therapy diminishes the leukemic load, and whether patients with low counts of CD26⁺CD35⁺ cells at diagnosis have a relatively higher risk of developing hematologic toxicity such as cytopenia during therapy. The relative levels of CD26⁺ and CD35⁺ stem cells for TFR will also be an important area of investigation.

Third, joint longitudinal analysis of leukemic as well as immune compartments is likely to be informative especially for TFR. The observations that patients with activated immune signature at diagnosis are more likely to optimally respond to TKIs (Radich et al., 2023 [DOI](#)) serve as apt reminder for collective analyses. Fourth, our capacity to generate single cell datasets, ironically outpaces our ability to extract information. Development of tools that allow biologists and clinicians to analyze large scale datasets without requiring dedicated bioinformatics infrastructure can overcome the challenge (Dhapola et al., 2022 [DOI](#)). Finally, the clonal relationships and lineage output of candidate BCR::ABL1⁺ LSC subpopulations remain unknown. By coupling sc-multiomics with barcoding analysis and lineage tracing (Wagner & Klein, 2020 [DOI](#)), it should be possible to evaluate, for instance, whether the BCR::ABL1⁺CD26⁺ cells detected in patients in TFR are a sub-clone that persists from diagnosis through therapy, and whether it is responsible for recurrence in TFR.

The notion of LSCs has been invoked for a long time in the genetically defined and molecularly targeted paradigm of CML (Holyoake & Vetrie, 2017 [DOI](#)). Still these entities remain essentially peripheral to the disease management (Zhao & Deininger, 2021 [DOI](#)). We posit that the heterogeneity of LSCs is a barrier towards their efficient measurement and safe purging. Our high-resolution single cell multiomics maps suggest how to probe and deconstruct heterogeneity of CML, thereby permitting inference of leukemic vs. non-leukemic cells, estimation of BCR::ABL1⁺ LSCs, enumeration of their molecular features, and prospective isolation. Understanding how the cellular heterogeneity and plasticity emerges in the absence of extensive genetic variability would inform if the fully leukemogenic residual cells could either be safely eliminated pharmacologically or kept perpetually suppressed by empowered immune system to avoid recurrence (Hsieh et al., 2021 [DOI](#); Zhao & Deininger, 2023 [DOI](#)). Alternatively, longitudinal sampling and sc-omics combined

with barcoding may also reveal if the stem cell subpopulations and states are interconvertible (Lenaerts et al., 2010 [DOI](#); Marjanovic et al., 2013 [DOI](#)), and therefore appear as potentially inexhaustible pool that can only be slowly eroded by several years of TKI treatment in a subset of patients. A more formidable barrier is the unstated but still presumed equivalence of immunophenotype and function. Rather than describing LSCs solely by surface markers, we propose that treating leukemic cellular clones as the fundamental units of selection and evolution during therapy would have a more meaningful impact in predicting response to existing therapy and switch to another TKI. The next line of advances will require assessing LSCs as pool of clones defined by their ability to contribute to primary and secondary resistance in patients on therapy, and recurrence in TFR without recourse to animal models.

Materials and methods

Patient samples

The bone marrow was obtained from patients enrolled in either BosuPeg clinical trial (NCT03831776), NordCML006 (NCT00852566), BFORE (NCT02130557), or from Skåne University Hospital, Lund after informed consent and processed according to the guidelines approved by regional research ethics committees of sites involved in the trial. The information on age, gender, TKIs, BCR::ABL1^{IS} % and therapy response as per ELN guidelines are provided as supplementary table 1. MNCs or CD34⁺ cells were enriched from the BM aspirates using magnetic microbeads and subsequently cryopreserved. The collection, processing and generation of CITE-seq of bone marrow from age matched healthy donors used in this study has been described previously (Sommarin et al., 2021 [DOI](#)).

CITE-seq sample preparation and FACS Sort

Sample preparation was performed according to (Stoeckius et al., 2017 [DOI](#)) with minor adaptations. In brief, CML MNC or CD34 enriched BM samples was thawed, FCR blocked 1/5 (130-046-703, Miltenyi) for 10 min and washed before staining. Samples were stained with a PeCy5 conjugated lineage cocktail (CD2, CD3, CD14, CD16, CD19, CD235a), CD34 FITC, and a subset of the CITE-seq panel for 30 min on ice. Samples were washed and split into two tubes and stained with either CD38-PeCy7 or CD38-CITE-seq antibody followed by individual hashtags and the rest of the CITE-seq panel for 30 min on ice (Supp. Table 2 and 3). Cells were washed and resuspended in PBS, 2 % FBS, 1/100 7AAD (Hyclone, 559925, BD Bioscience). Two populations (Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-low}) were sorted per sample using FACSARIAII/Aria III (BD Bioscience) into 300 µl PBS, 0.04 % BSA. Sorted cells were centrifuged at 300 x g for 10 min and volume was adjusted to 45 µl.

CITE-seq library preparation and sequencing

CITE-seq library preparation and sequencing Cells were processed through Chromium Controller (10x Genomics) with a total of 20-30.000 cells loaded per lane. Single-cell cDNA, HTO (Hashtags) and ADT (CITE-seq antibodies) libraries was prepared using Chromium Single Cell 3' V3 as per manufacturer instructions (CG000183 Rev C) as reported previously (Stoeckius et al., 2017 [DOI](#)). In brief, to increase HTO and ADT library yield, HTO and ADT specific primers were spiked-in during the cDNA amplification step. cDNA and HTO/ADT libraries were isolated using 2X SPRI beads per manufacturers protocol. HTO and ADT libraries were subjected to adaptor ligation and sample index in separate PCRs using KAPA Hifi PCR Readymix (10 and 11 PCR cycles respectively). The cDNA library was subjected to fragmentation, end repair, A-tailing, adaptor ligation and sample index PCR (12 cycles). cDNA, HTO and ADT libraries were sequenced together (65 % cDNA, 30 %ADT, 5 % HTO) on Illumina sequencers with the following read length configuration: Read1=28, i7=8, i5=0, Read2=91. The raw data was processed using Cell Ranger 3.0.2 with GRCh38 as a reference genome.

CITE-seq analysis

Cell-gene matrices produced by Cell Ranger were analyzed using Scarf (v.0.18.12) (Dhapola et al., 2022 [DOI](#)). In brief, cells with low or high gene count (<1000 or >9000 genes) and cells with high percentage mitochondrial gene UMIs (>11 % or >6 % for CML samples and healthy BM reference, respectively) were excluded from the analysis. The matrices were demultiplexed using Otsu thresholding (automatized pixel-based thresholding of HTO count per cell, background buffer = 0.1, override value = 0.5) (Otsu thresholding was run twice for CML4, CML9. For nBM the thresholds were adjusted to HTO_1 = 32, HTO_2 = 18, HTO_3 = 17). HTO UMAPs were generated through Scarf (dims = 0) and clustered with Paris clustering (n_clusters = 2). If any RNA leiden cluster overlapped with HTO paris cluster > 70 % and conversely, it was considered doublets and excluded from analysis.

Post-demultiplexing, for each sample, 2000 highly variable genes (HVGs) were identified (min_cells = 50) with the “mark_hvgs” function in Scarf and used for PCA (principal component analysis) (CML8; HVGs = 1000, min_cells = 30). KNN (K-Nearest Neighbors) graphs were calculated based on the top 20 principal components (n_neighbours = 11) and utilized to build UMAP embedding of the cells (min_dist=1, spread=2, n_epochs= 2000). Clustering was performed using Lieden clustering (resolution set to 1 and 0.76 for CML and nBM samples respectively).

Projection of Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} CML subpopulations

Mapping reference

The nBM is merged CITE-seq data of two Lin⁻CD34⁺ FACS sorted bone marrow samples (Sommarin et al., 2021 [DOI](#)) reanalyzed as described above’. Cluster identity was determined by (a) the top 10 genes in “run_marker_search” function in Scarf. In brief, each genes expression value is ranked per cell and a gene’s mean rank per cluster is divided by the sum of mean to determine the cluster specificity, (b) CLR (centred-log ratio) normed ADT expression across cluster was used to determine the immunophenotypic identity the cells, and (c) cell cycle scoring using the “run_cell_cycle_scoring” function in Scarf was used to estimate cell cycle phase.

Label transfer through reference-based cell projection

CML subpopulations were mapped to Lin⁻CD34⁺ nBM using Scarf (v.0.18.12). In short, cells were projected using ‘K-Nearest Neighbours (KNN) mapping through Scarf’s “run_mapping” function (neighbours = 5). Cell cluster identity for CML cells was determined using the “get_target_classes” function with threshold set to 0.5 (> 50 % of the total weight score from the five top matched reference cells must be cluster specific to assign identity). Mapping scores (how frequently the reference cells end up as one of top 5 neighbours of the projected CML cells) for the individual reference cells was calculated per projected sample using “get_mapping score”. To visualize the mapping score and compare the mapping across samples, cells size of reference cells was set proportional to the score in the healthy bone marrow reference UMAP.

Following mapping of Lin⁻CD34⁺CD38^{-/low} CML cells (n = 8), the primitive clusters from all patients were merged in Scarf. The “make_graph” function in Scarf was used to identify the primitive CMLs cells top 500 HVGs (min_cells =30, gender specific genes excluded). Lin⁻CD34⁺CD38^{-/low} healthy bone marrow cells (n = 1) were projected onto the healthy reference as described above and the primitive cluster was added to the primitive CML cell clusters in Scarf. The UMAP was generated using the 500 CML specific HVGs (k =11, PC = 20, n_epoch = 500) and with PCA fitted to the primitive CML cells only using Scarf’s “make graph” function. Subsequently, cells were clustered using Leiden clustering method (resolution = 0.1).

BCR::ABL1 status of the primitive clusters was determined using two DEG signatures: (1) Normal HSC vs BCR::ABL1⁺, and (2) BCR::ABL1⁻ vs BCR::ABL1⁺ where Lin⁻CD34⁺CD38^{-/low}BCR::ABL1⁺ cells have been validated through single-cell RT-qPCR (Giustacchini et al., 2017). Up- and down-regulated genes were defined as log2 fold change > 1 and < 0, respectively. The expression of signature genes was normalized in Scarf, Z-scored to provide an average value count of the signature per cell and visualized in the UMAP using Scarf.

To visualize the BCR::ABL1⁺ primitive cells alongside the full CML heterogeneity, Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-/low} CML CITE-seq data from the same patient sample was merged in Scarf (The sub populations were FACS sorted at the time point and sequenced together). UMAPs of the merged data sets were generated as previously described (2000 HVGs, neighbours = 11, PC = 20). Cell indexes from BCR::ABL1⁺ and BCR::ABL1⁻ primitive CML clusters was matched to the Lin⁻CD34⁺CD38^{-/low} cell data.

ADT expression of BCR::ABL1⁺ and BCR::ABL1⁻ cells were retrieved from the zarr file in Scarf. The data thereafter was CLR (centered-log ratio) normalized and the log2 fold change was calculated between the two groups of cells using mean expression. P-values were calculated with Mann-Whitney U test and they were corrected for multiple hypotheses testing using Bonferroni method.

Analysis of differentially expressed genes

DESeq2 was used to identify differentially expressed genes. Significantly up- or down-regulated was selected based on adjusted p-value < .01 and log2 fold change > 1 or < -1. Subsequently, volcano plots showing log2 fold change and -Log10 p-values were generated. Mitochondrial/ribosomal genes, as well as genes with a total count below 10 was excluded from the analysis.

To compare gene expression between clusters 1 to 11 from Lin⁻CD34⁺ CML cells, n = 9) and their normal counterparts, their corresponding normal bone marrow reference clusters (n = 2) were randomly pseudo bulked into 3 replicates. Cluster specific signature genes were defined as significant up- or down-regulated genes unique to the cluster comparison, and not shared with not any other clusters. CML signature genes were defined as significantly up- or down-regulated genes present in all 11 clusters. Heatmap was generated from the DESeq2 data output (VST), showing the log2 mean expression of all samples per cluster. BCR::ABL1⁺ and BCR::ABL1⁻ clusters > 25 cells was used for DEG analysis of Lin⁻CD34⁺CD38^{-/low} primitive cells (n = 7).

ADT gating

ADT expression was CLR (*centered-log ratio*) normalized in Scarf. The normalized data was transformed into its antilog, multiplied with a scaling factor of 1000 and exported as FCS files. The FCS files was analysed using FlowJo software. Gated cells ADT scale values were exported as CSV files and matched to cells ADT expression in CITE-seq data to determine original cell identity.

Gene expression deconvolution

The gene expression counts from CITE-seq molecularly defined populations served as input for CIBERSORTx (Newman et al., 2019) at default settings run on its docker version. Deconvolution of publicly available CD34⁺ bulk gene expression microarray dataset GSE14671 (McWeeney et al., 2010) was performed after robust multichip average (RMA) normalization (Irizarry et al., 2003). Upon estimation of relative abundances of subpopulations from CD34⁺ cells, the values were normalized to 1, the value for each subpopulation represented its percentage/fraction within the query sample. Individual patients were annotated as non-responders or responders to Imatinib therapy as defined in the original study (McWeeney et al., 2010); non-responders with >65% Ph⁺ metaphases (the patients did not achieve even a minor cytogenetic response as described by the original authors), responders with 0% with Ph⁺ metaphases (achieved CCyR) after

12 months of Imatinib therapy. Notably, the threshold for defining therapy response was proposed by the original authors and the patients were labeled as either responder or non-responder (McWeeney et al., 2010 [DOI](#)). We adhered to the patient labels provided by the original authors as described in GSE14671.

Flow cytometry for phenotyping

Lymphoprep kits were used to separate mononuclear cells (MNCs), and magnetic microbeads were used to enrich CD34⁺ cells (Miltenyi). Antibodies against lineage-specific markers and those for specific populations mentioned in Supplementary Table 3 were used to stain the cells. Sorting and analysis were done using FACS ARIAII/III or LSR FORTRESSA (BD Biosciences); data analysis was done with FlowJo (Tree Star)

qPCR for BCR::ABL1

RNA from FACS sorted cells was purified using Single Cell RNA Purification Kit (Norgen, cat# 51800) followed by cDNA synthesis using High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, cat# 4368814). Following pre-amplification of cDNA (SsoAdvanced PreAmp Supermix (BioRad, cat# 172-5160) for 12 cycles, qPCR was performed with TaqMan gene expression master mix (Applied Biosystems, cat# 4369016) and TaqMan probes for BCR::ABL1 (Assay ID: Hs03024541_ft), GAS2 (Assay ID: Hs00169477_m1 and GAPDH (Assay ID: Hs02758991_ft).

Time to first division in vitro

Lin⁻CD34⁺CD38^{-/low}CD45RA⁻CD26⁺CD35⁻ or CD26⁻CD35⁺ BM single cells from two CML patients were sorted into individual wells of 96-well U shaped-bottom TPP plates using the FACS Aria III flow cytometer (Becton Dickinson). Cells were sorted directly into 200 µl Stem Span Serum-Free Expansion Medium (SFEM, StemCell Technologies), supplemented with 100 units/ml penicillin, 0.1 mg/ml streptomycin (Hyclone) and the following cytokines from PeproTech: SCF (100 ng/ml), Flt3L (100 ng/ml), TPO (50 ng/ml), and IL7 (10 ng/ml). Cells were visualized and counted in each well thrice a day using an inverted microscope for >140 hours. Dead cells as well as cells that did not divide during this period were excluded from the analysis. The Curve fit-sigmoid approach in GraphPad Prism was used to analyze the data.

Data availability

The data from sequencing have been submitted to GEO under accession ID: GSE236233 and are publicly available as of the date of publication. Previously published data used within this study is available at: GSE173076 (nBM), GSE1467 (deconvolution).

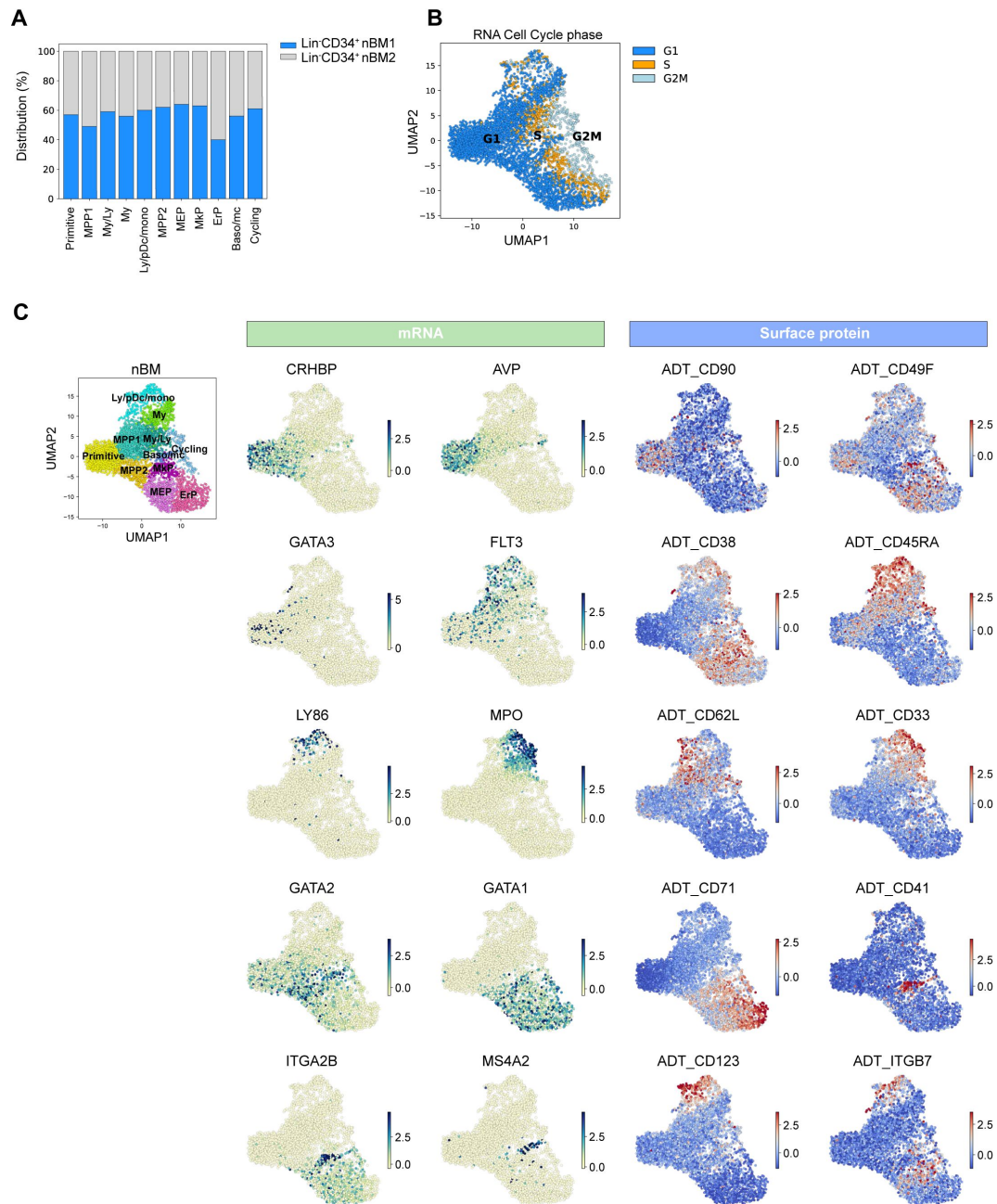
Author contributions

RW, GK, and RKT conceived and designed the study, that was supervised by GK; RW designed the CITE-seq ADT panel with inputs from MNES, LGU, RKT, and GK. CITE-seq experiments were performed by RW with help from LGU; FACS, and qPCR analysis were done by LGU, in vitro cell division kinetics measurements were recorded by LGU and FS with contribution from GK, RKT and RW. CITE-seq analysis was done by RW with assistance from PD, bulk gene expression deconvolution analysis was implemented by SS with assistance from GK and RKT. JR, HHJH, and SM provided patient material and data from the clinical trials as well as interpretation of data and advice. RW summarized the data and generated the figures. RW, RKT and GK wrote the manuscript with input from all authors. All authors read and approved the final version.

Conflict of interest disclosure

GK and PD are board members and have equity in Nygen Analytics AB, JR reports receiving honoraria and research funding from Novartis and Bristol-Myers Squibb (BMS) and honoraria from Ariad. HHjH has received honoraria from Pfizer, Novartis, BMS, and Incyte. SM has received honoraria and research funding from BMS, and research funding from Novartis, Janpix, and honoraria from Dren Bio. Sample collection from patients in the BosuPeg and BFORE clinical trials was supported by Pfizer investigator grant. Other honoraria, and research funds were for projects unrelated to this study. The remaining authors declare no conflicts of interest.

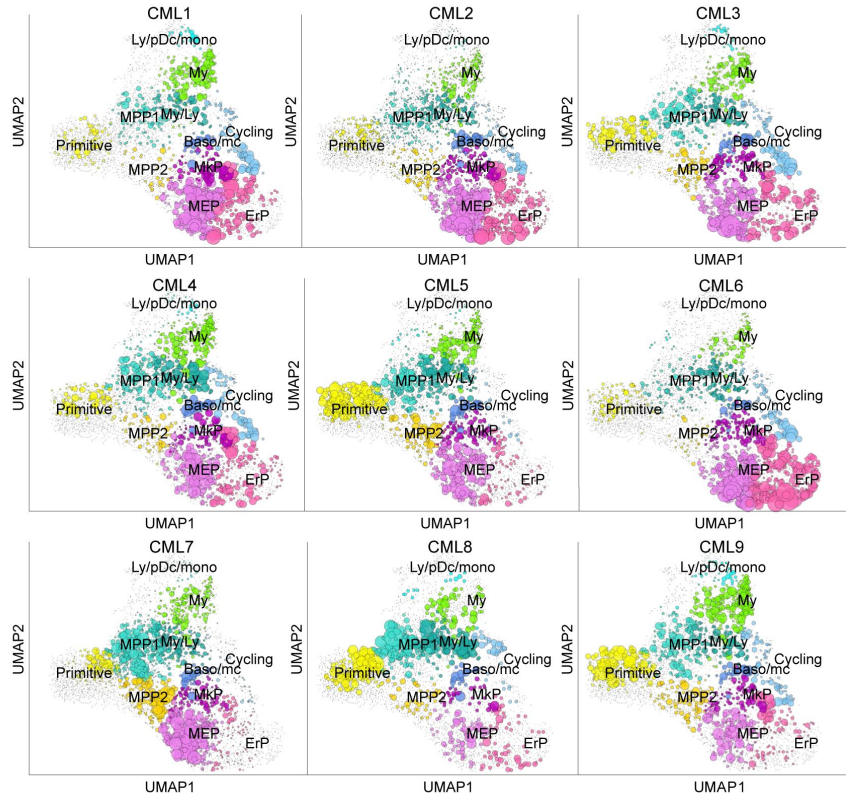
Supplementary Figures



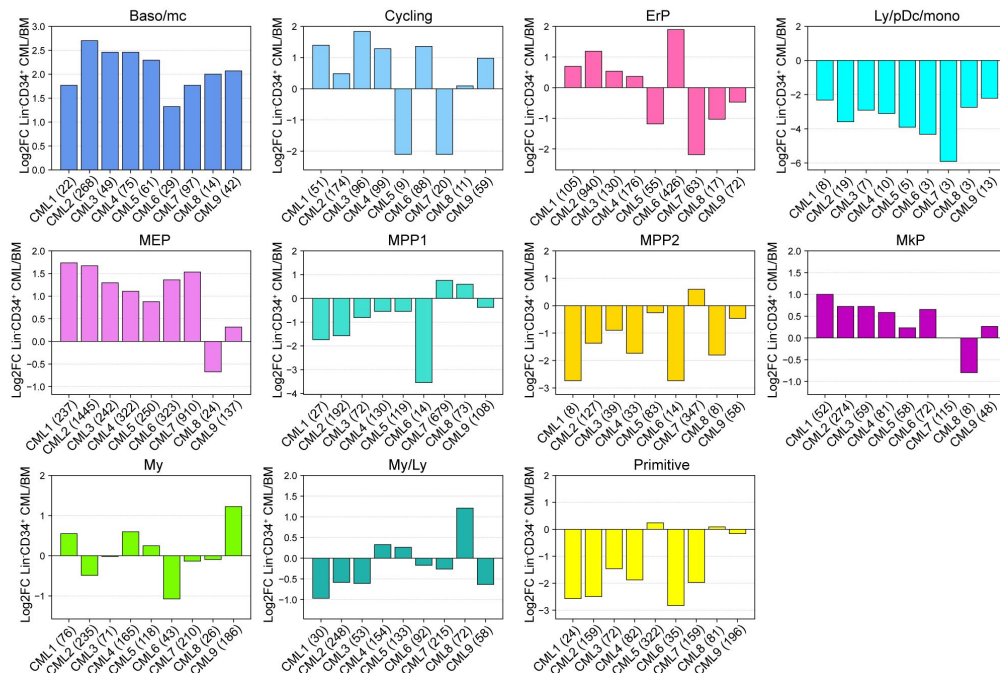
Supplementary Figure 1.

- (A) Distribution of annotated cell clusters across the two Lin-CD34+ nBM samples.
- (B) Cell cycle scoring: G1, S, G2/M of UMAP embedded Lin-CD34+ cells from nBM using Scarf.
- (C) Visualization of the relative mRNA (left panels) and ADT expression (right panels) of a selection of genes and surface markers within the Lin-CD34+ nBM reference.

A



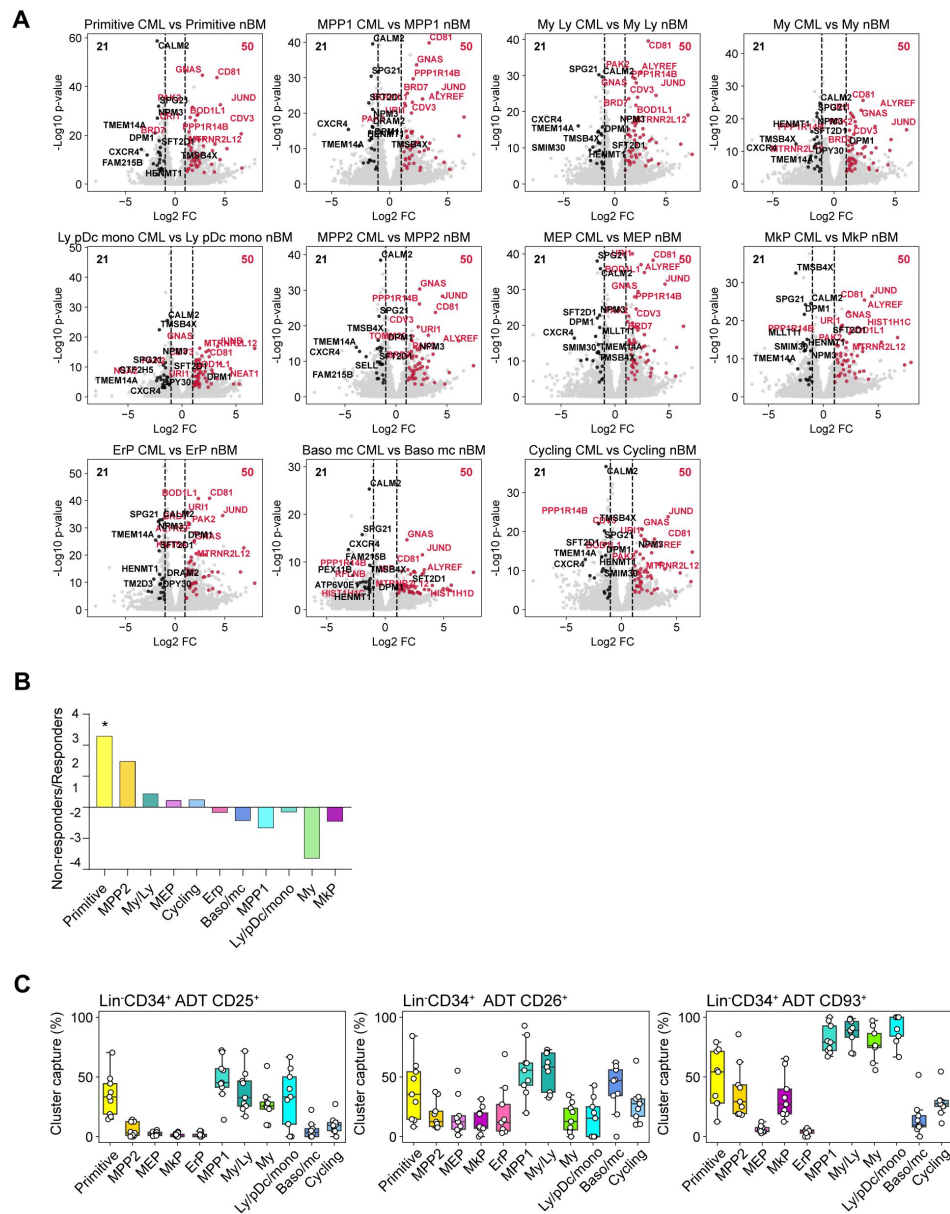
B



Supplementary Figure 2.

(A) UMAP plots showing the mapping score for the individual Lin⁻CD34⁺ CML samples at diagnosis when mapped onto the nBM reference; mapping was performed using Scarf (Dhapola et al., 2022). Circle size indicates mapping score and circle color specify cluster identity of the reference cell.

(B) Bar plots showing the log2 fold change in cluster distribution (%) within the CML Lin⁻CD34⁺ compartment at diagnosis when compared to nBM for the individual CML patients.

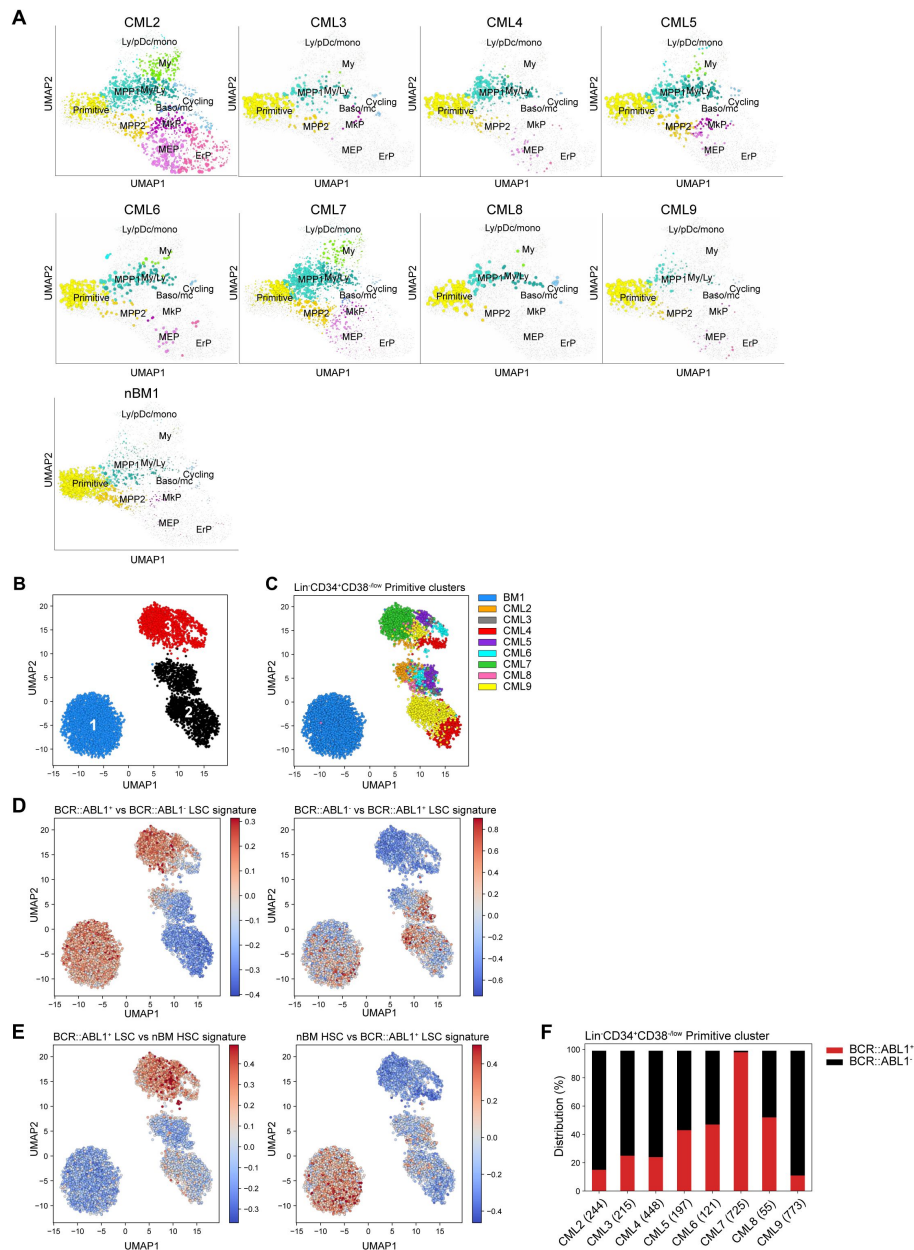


Supplementary Figure 3.

(A) Volcano plots showing differentially expressed genes between Lin⁺CD34⁺ CML ($n = 9$) at diagnosis and nBM cells across clusters (DESeq2). Red and black dots represent genes found to be consistently significant up- or down-regulated across all cluster comparisons in CML, respectively (adjusted p -value $< .01$, $\log_2$ fold change $= >1 / < -1$). Vertical dotted line marks a $\log_2$ fold change equal to 1 and -1 .

(B) The fold change in proportions for cell clusters between all non-responders and responders (CD34⁺ cells isolated from both PB and BM); statistical significance shown by asterisk *; student t -test, p -value $< .05$.

(C) Assessment of the CML related ADTs; CD25, CD26, CD93 capacity to capture primitive cells, and specific progenitors across patients by gating on their expression within the Lin⁺CD34⁺ compartment (x-axis shows specific clusters, y-axis: percentage of specific cluster captured; each white circle represents a patient sample).



Supplementary figure 4.

(A) UMAP plots showing the mapping score for individual Lin⁻CD34⁺CD38^{-low} CML samples (CML2-8) at diagnosis and Lin⁻CD34⁺CD38^{-low} normal BM when mapped onto the Lin⁻CD34⁺ nBM reference; mapping was performed using Scarf. Circle size indicates mapping score and circle color specify cluster identity of the reference cell.

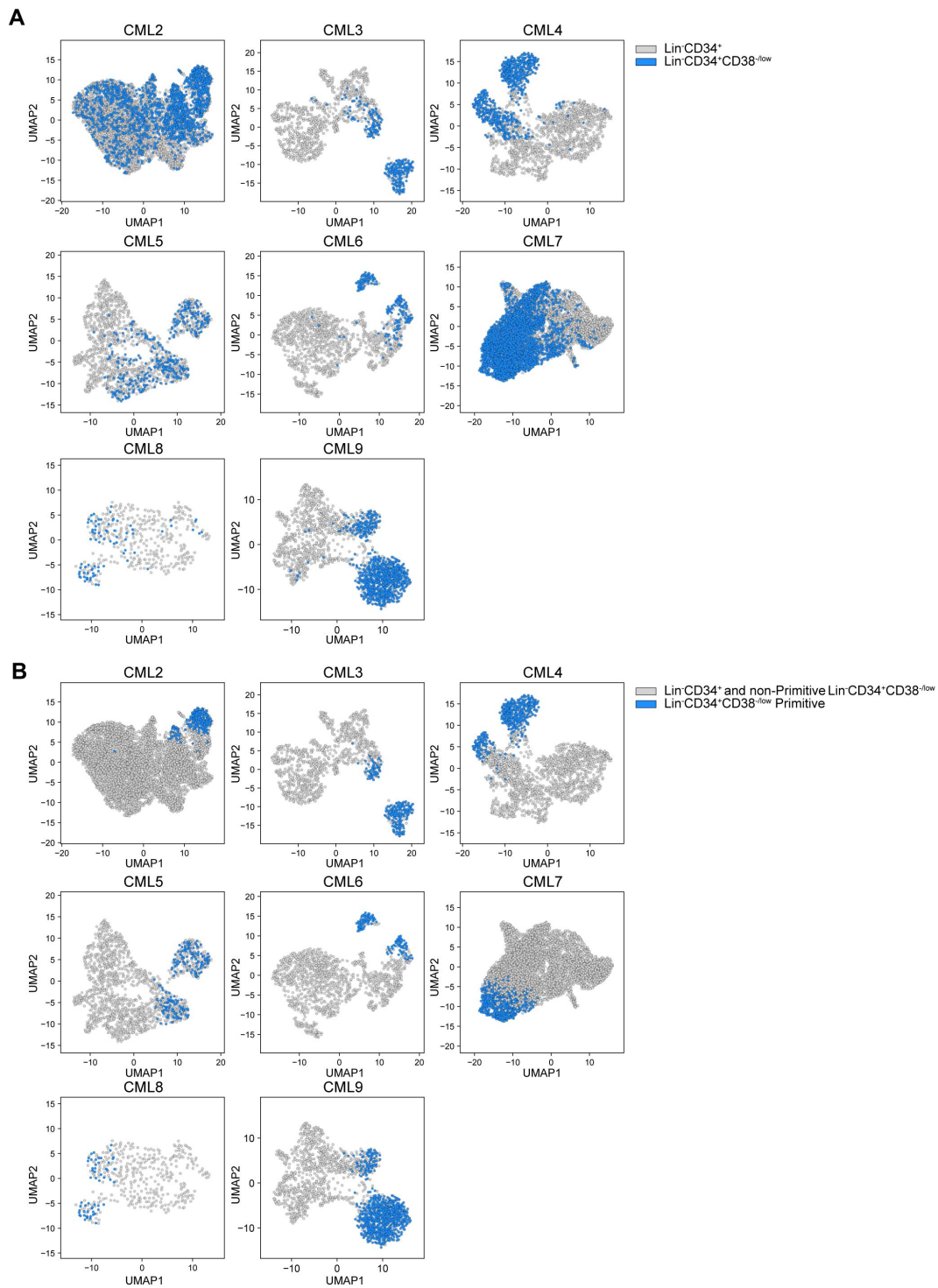
(B) UMAP embedding of Lin⁻CD34⁺CD38^{-low} primitive cluster CML (n = 8, a total of 2778 cells) and normal BM cells showing clusters following Leiden clustering analysis.

(C) UMAP embedding of Lin⁻CD34⁺CD38^{-low} primitive cluster CML (n = 8, a total of 2778 cells) at diagnosis and normal BM cells where cells are colored by sample identity.

(D) Same UMAP as in B-C above, showing the relative expression of BCR::ABL1⁺ LSC (left panel) and BCR::ABL1⁻ LSC signature genes (right panel) used to define BCR::ABL1⁺ primitive cells (Signature genes are from DEG analysis comparing CP-CML Lin⁻CD34⁺CD38^{-low} BCR::ABL1⁺ and BCR::ABL1⁻ (Giustacchini et al., 2017)).

(E) Same UMAP as in B-C above, showing the relative expression of BCR::ABL1⁺ LSC (left panel) and nBM HSC signature genes (right panel) used to define BCR::ABL1⁺ primitive cells (Signature genes are from DEG analysis comparing CP-CML Lin⁻CD34⁺CD38^{-low} BCR::ABL1⁺ and nBM Lin⁻CD34⁺CD38^{-low} cells (Giustacchini et al., 2017)).

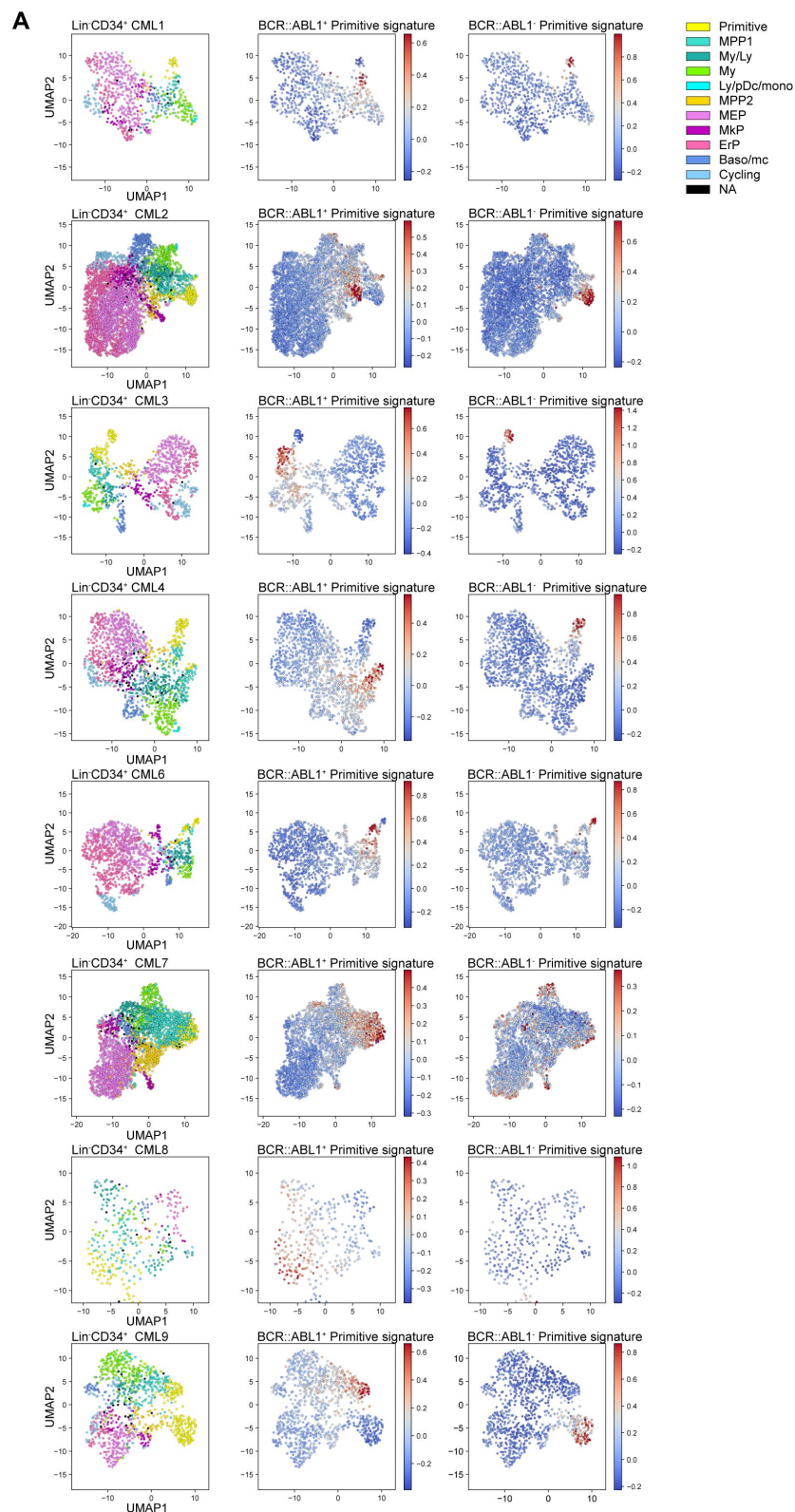
(F) Quantification of Lin⁻CD34⁺CD38^{-low} primitive BCR::ABL1⁺ (red) and BCR::ABL1⁻ (black) cells per CML patient at diagnosis. Total primitive cells per patient is indicated at the bottom.



Supplementary figure 5.

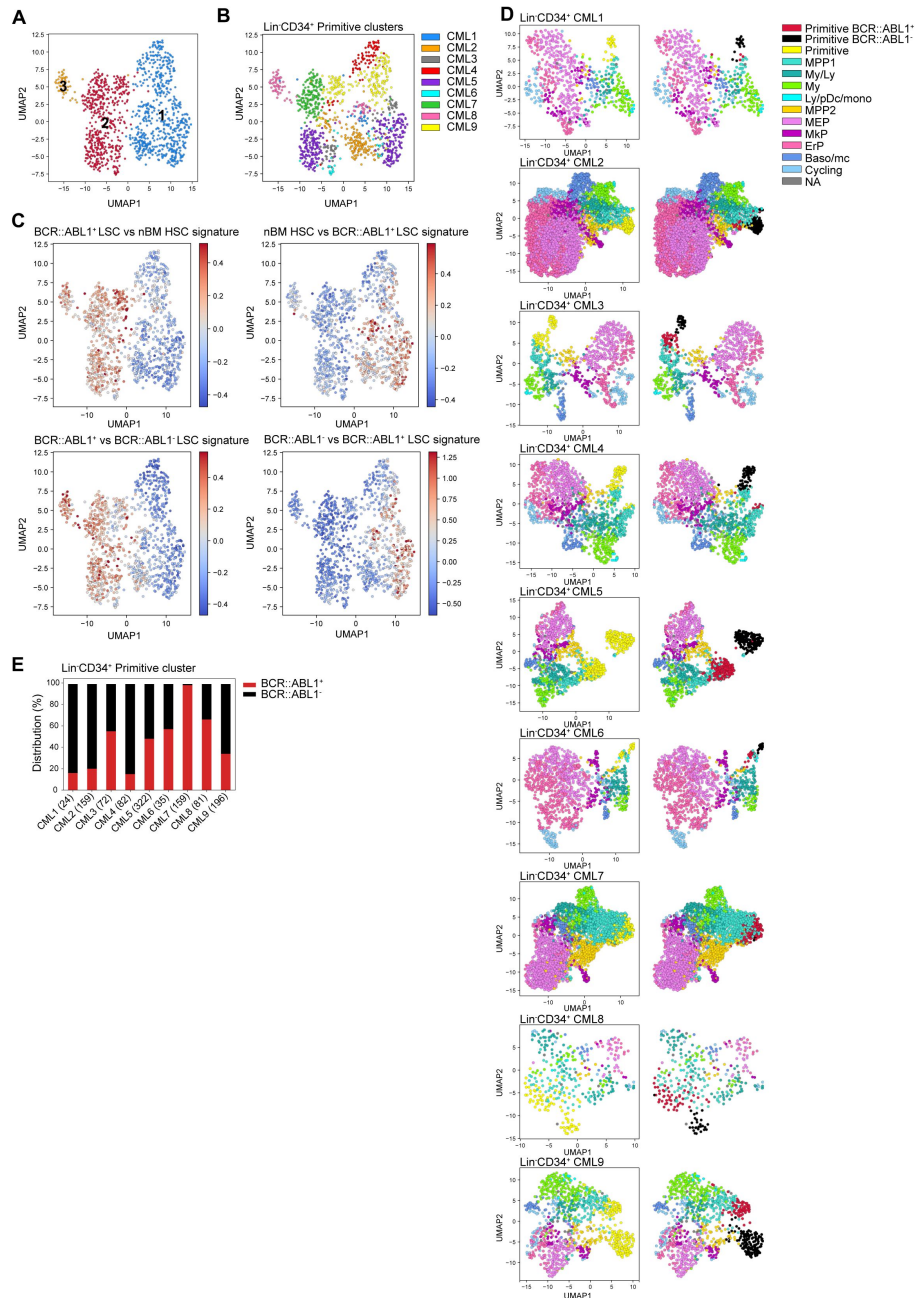
(A) UMAP plots where CITE-seq data from Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-low} sorted cells from the same CML patients have been merged, and Lin⁻CD34⁺CD38^{-low} cells are highlighted in blue.

(B) UMAP plots where CITE-seq data from Lin⁻CD34⁺ and Lin⁻CD34⁺CD38^{-low} sorted cells from the same CML patients have been merged, where Lin⁻CD34⁺CD38^{-low} primitive annotated cells are highlighted in blue.



Supplementary figure 6.

(A) UMAP plots of Lin⁻CD34⁺ cells from CML patients 1-4 and 6-9, showing cells colored according to cluster identity given after mapping to nBM (left panel); the relative expression of Lin⁻CD34⁺CD38^{low} Primitive BCR::ABL1⁺ signature genes (center panel) and the relative expression of Lin⁻CD34⁺CD38^{low} Primitive BCR::ABL1⁻ signature genes (right panel). DEG analysis is shown in **Figure 4C** and relative expression of signatures for CML5 is found in **Figure 4D**. (red = high expression, blue = low expression).



Supplementary figure 7.

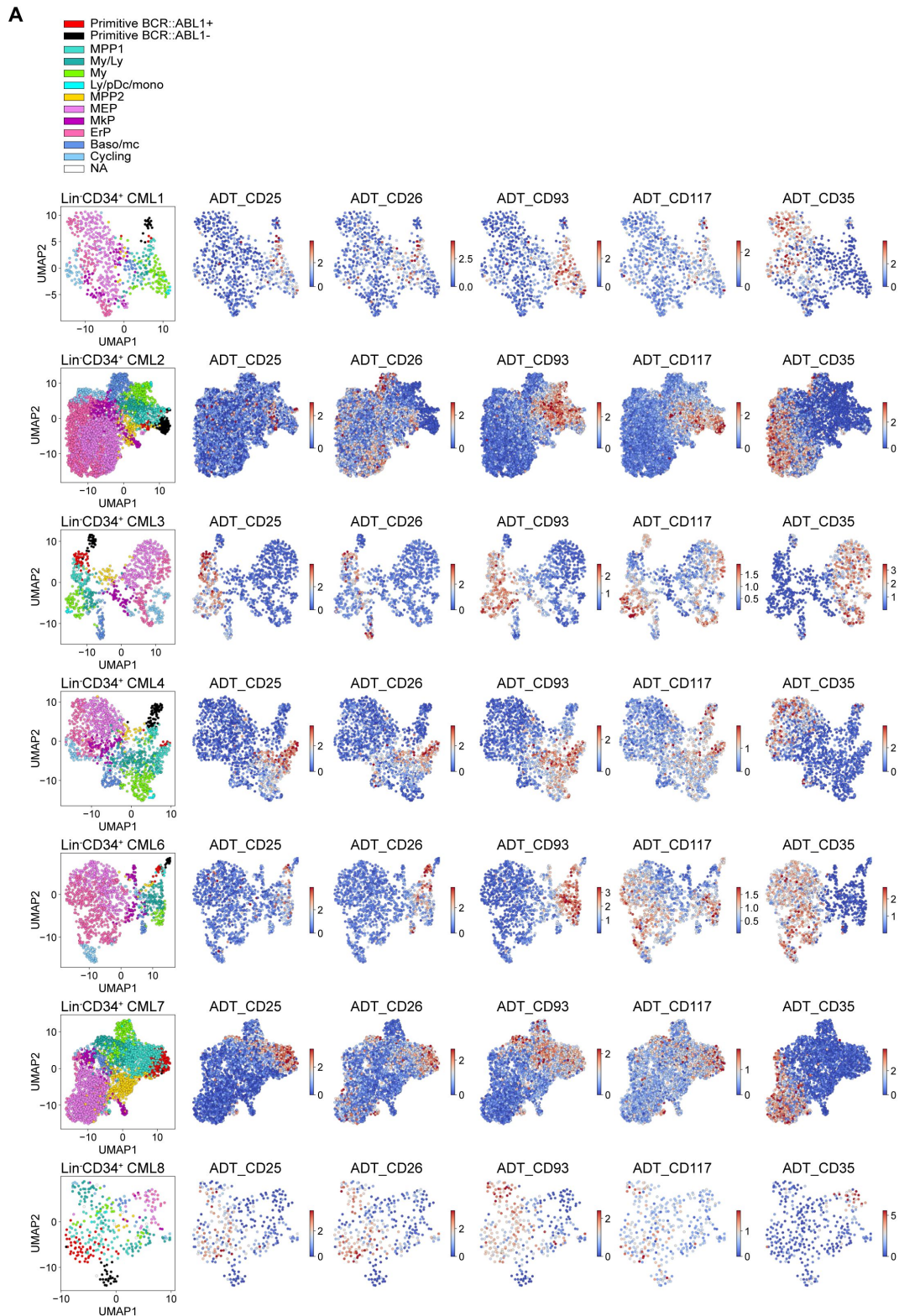
(A) UMAP plots of CML Lin⁻CD34⁺ primitive cluster cells from all CML patients (n = 9, a total of 1 106 cells) showing clusters following Leiden clustering analysis.

(B) UMAP plots of CML Lin⁻CD34⁺ primitive cluster from all CML patients (n = 9, a total of 1 106 cells) where the cells are colored as per the sample identity

(C) UMAP plots of CML Lin⁻CD34⁺ primitive cluster cells from all CML patients showing the relative expression of two DEG signatures used to define BCR::ABL1⁺ primitive cells: (1) BCR::ABL1⁺ LSC vs nBM HSC signature genes (top row, left panel = up-regulated genes in BCR::ABL1⁺ LSCs, right panel = genes up-regulated in nBM HSCs), (2) BCR::ABL1⁺ LSC vs BCR::ABL1⁻ LSC (bottom row, left panel = genes up-regulated in BCR::ABL1⁺ LSC, right panel = genes up-regulated in BCR::ABL1⁻ LSC) (Giustacchini et al., 2017 [\[1\]](#)).

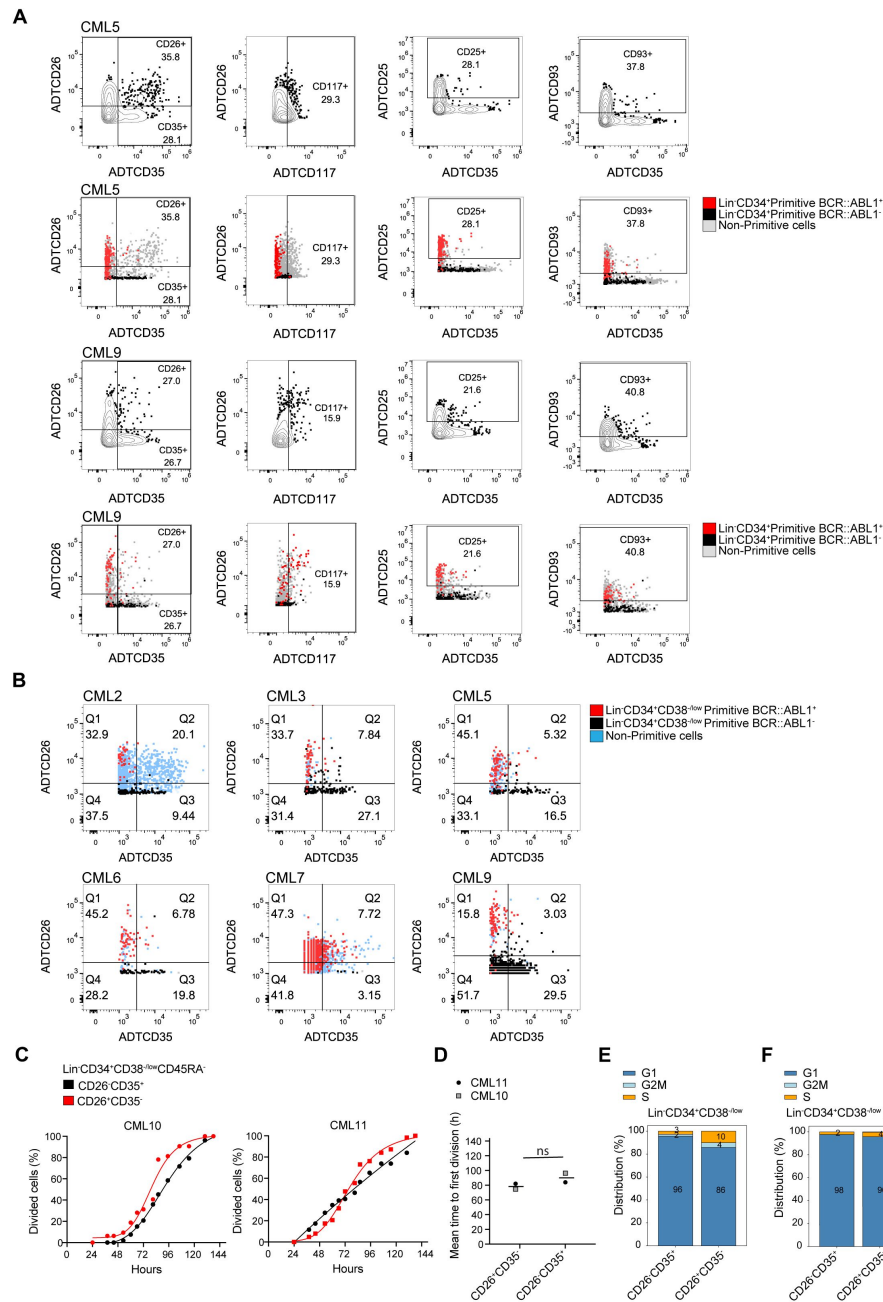
(D) UMAP plots of Lin⁻CD34⁺ cells from individual CML patients (1-9) showing primitive cluster cells (yellow cluster, left panel) annotated as BCR::ABL1⁺ (red) and BCR::ABL1⁻ (black)(right panel).

(E) Quantification of Lin⁻CD34⁺ primitive BCR::ABL1⁺ (red, cluster 2-3 **Supp. Fig. 7A** [\[1\]](#)) and BCR::ABL1⁻ (black, cluster 1 **Supp. Fig. 7A** [\[1\]](#)) cells per CML patient at diagnosis. Total primitive cells are indicated at the bottom.



Supplementary figure 8.

(A) UMAP plots of Lin⁻CD34⁺ cells from CML patients 1-4 and 6-8 showing the relative ADT expression of a selection of markers (CD25, CD26, CD93, CD117, CD35) (red = high expression, blue = low expression). ADT expression for CML patient 5 and 9 is shown in [Fig. 5B](#).



Supplementary figure 9.

(A) Plots showing representative gates on ADT expression for CD26, CD35, CD117, CD25, CD93 within the Lin⁻CD34⁺ CITE-seq data using FlowJo. Gates are shown for CML5 and CML9.

(B) Plots showing the gating on CD26 and CD35 ADT expression within the Lin⁻CD34⁺CD38^{-low} CITE-seq data for CML patients 2-3, 5-7 and 9 using FlowJo. Plots for CML patient 4 and 8 is shown in **Fig. 6D**.

(C) Cumulative plots of time to first division kinetics from Lin⁻CD34⁺CD38^{-low} CD45RA⁻CD26⁺CD35⁻ and CD26⁻CD35⁺ single cells for two individual CML patients at diagnosis (n = 2, CML patients 10 and 11). Dead cells and non-dividing cells were excluded from the analysis.

(D) Mean time to first division (hours) for Lin⁻CD34⁺CD38^{-low}CD45RA⁻CD26⁺CD35⁻ or CD26⁻CD35⁺ single cells from the two CML patients shown in C (n = 2, CML patients 10 and 11) (p-value = ns., .84; paired t-test).

(E) Mean cell cycle status of ADT gated CD26⁺CD35⁻ or CD26⁻CD35⁺ cells, after running Scarf's cell cycle scoring on the Lin⁻CD34⁺CD38^{-low} CITE-seq data for patients individually (n = 8, CML2-9).

(F) Mean cell cycle status of ADT gated CD26⁺CD35⁻ or CD26⁻CD35⁺ cells, after running Scarf's cell cycle scoring on Lin⁻CD34⁺CD38^{-low} Primitive cluster for patients individually (n = 8, CML2-9).

Supplementary tables

Supplementary Table 1

CML sample cohort

Supplementary Table 2

CITE-seq antibodies

Supplementary Table 3

FACS antibodies

Supplementary Table 4

List of marker genes for nBM clusters

Supplementary Table 5

List of DEG uniquely changed per cluster in CML vs. nBM clusters 1-11 along with their fold change.

Supplementary Table 6

List of DEG up- and down-regulated in all clusters in CML vs. nBM clusters 1-11.

Supplementary Table 7

Lits of DEG along with their fold change between BCR::ABL1⁺ LSCs versus BCR::ABL1⁻ stem cells.

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Editors

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Reviewer #1 (Public Review):

Summary:

This manuscript by Warfvinge et al. reports the results of CITE-seq to generate single-cell multi-omics maps from BM CD34+ and CD34+CD38- cells from nine CML patients at diagnosis. Patients were retrospectively stratified by molecular response after 12 months of TKI therapy using European Leukemia Net (ELN) recommendations. They demonstrate heterogeneity of stem and progenitor cell composition at diagnosis, and show that compared to optimal responders, patients with treatment failure after 12 months of therapy demonstrate increased frequency of molecularly defined primitive cells at diagnosis. These results were validated by deconvolution of an independent previously published dataset of bulk transcriptomes from 59 CML patients. They further applied a BCR-ABL-associated gene signature to classify primitive Lin-CD34+CD38- stem cells as BCR:ABL+ and BCR:ABL-. They identified variability in the ratio of leukemic to non-leukemic primitive cells between patients, showed differences in expression of cell surface markers and determined that a combination of CD26 and CD35 cell surface markers could be used to prospectively isolate the two populations. The relative proportion of CD26-CD35+ (BCR:ABL-) primitive stem cells was higher in optimal responders compared to treatment failures, both at diagnosis and following 3 months of TKI therapy.

Strengths:

The studies are carefully conducted and the results are very clearly presented. The data generated will be a valuable resource for further studies. The strengths of this study are the application of single-cell multi-omics using CITE-Seq to study individual variations in stem and progenitor clusters at diagnosis that are associated with good versus poor outcomes in response to TKI treatment. These results were confirmed by deconvolution of a historical bulk RNAseq data set. Moreover, they are also consistent with a recent report from Krishnan et al. and are a useful confirmation of those results. The major new contribution of this study

is the use of gene expression profiles to distinguish BCR-ABL+ and BCR-ABL- populations within CML primitive stem cell clusters and then applying antibody-derived tag (ADT) data to define molecularly identified BCR:ABL+ and BCR-ABL- primitive cells by expression of surface markers. This approach allowed them to show an association between the ratio of BCR-ABL+ vs BCR-ABL- primitive cells and TKI response and study dynamic changes in these populations following short-term TKI treatment.

Weaknesses:

The number of samples studied by CITE-Seq is limited. However, the authors have confirmed their key observations in additional samples. The BCR-ABL+ versus BCR-ABL- status of cells was not confirmed by direct sequencing for BCR-ABL. However, we recognize that the methodologies to perform these analyses on single cells is still evolving and the authors have shown that CD26 and CD35 expression can consistently identify BCR-ABL+ versus BCR-ABL- cells. It will be of interest to learn whether the GEP and surface markers identified here can distinguish BCR-ABL+ primitive stem cells later in the course of TKI treatment.

<https://doi.org/10.7554/eLife.92074.2.sa1>

Reviewer #3 (Public Review):

Summary:

In this study, Warfvinge and colleagues use CITE-seq to interrogate how CML stem cells change between diagnosis and after one year of TKI therapy. This provides important insight into why some CML patients are "optimal responders" to TKI therapy while others experience treatment failure. CITE-seq in CML patients revealed several important findings. First, substantial cellular heterogeneity was observed at diagnosis, suggesting that this is a hallmark of CML. Further, patients who experienced treatment failure demonstrated increased numbers of primitive cells at diagnosis compared to optimal responders. This finding was validated in a bulk gene expression dataset from 59 CML patients, in which it was shown that the proportion of primitive cells versus lineage-primed cells correlates to treatment outcome. Even more importantly, because CITE-seq quantifies cell surface protein in addition to gene expression data, the authors were able to identify the BCR/ABL+ and BCR/ABL- CML stem cells express distinct cell surface markers (CD26+/CD35- and CD26-/CD35+, respectively). In optimal responders, BCR/ABL- CD26-/CD35+ CML stem cells were predominant, while the opposite was true in patients with treatment failure. Together, these findings represent a critical step forward for the CML field and may allow more informed development of CML therapies, as well as the ability to predict patient outcomes prior to treatment.

Strengths:

This is an important, beautifully written, well-referenced study that represents a fundamental advance in the CML field. The data are clean and compelling, demonstrating convincingly that optimal responders and patients with treatment failure display significant differences in the proportion of primitive cells at diagnosis, and the ratio of BCR-ABL+ versus negative LSCs. The finding that BCR/ABL+ versus negative LSCs display distinct surface markers is also key and will allow for more detailed interrogation of these cell populations at a molecular level.

Weaknesses:

CITE-seq was performed in only 9 CML patient samples and 2 healthy donors. Additional samples would greatly strengthen the very interesting and notable findings.

Author Response

The following is the authors' response to the original reviews.

eLife assessment

This study, utilizing CITE-Seq to explore CML, is considered a useful contribution to our understanding of treatment response. However, the reviewers express concern about the incomplete evidence due to the small sample size and recommend addressing these limitations. Strengthening the study with additional patient samples and validation measures would enhance its significance.

We thank the editors for the assessment of our manuscript. In view of the comments of the three reviewers, we have increased the number of CML patient samples analyzed to confirm all the major findings included in the manuscript. In total, more than 80 patient samples across different approaches have now been analyzed and incorporated in the revised manuscript.

To the best of our knowledge, this is the first single cell multiomics report in CML and differs substantially from the recent single cell omics-based reports where single modalities were measured one at a time (Krishnan et al., 2023; Patel et al., 2022). Thus, the sc-multiomic investigation of LSCs and HSCs from the same patient addresses a major gap in the field towards managing efficacy and toxicity of TKI treatment by enumerating CD26+CD35- LSCs and CD26-CD35+ HSCs burden and their ratio at diagnosis vs. 3 months of therapy. The findings suggest design of a simpler and cheaper FACS assay to simultaneously stratify CML patients for TKI efficacy as well as hematologic toxicity.

Reviewer 1:

Summary:

This manuscript by Warfvinge et al. reports the results of CITE-seq to generate singlecell multi-omics maps from BM CD34+ and CD34+CD38- cells from nine CML patients at diagnosis. Patients were retrospectively stratified by molecular response after 12 months of TKI therapy using European Leukemia Net (ELN) recommendations. They demonstrate heterogeneity of stem and progenitor cell composition at diagnosis, and show that compared to optimal responders, patients with treatment failure after 12 months of therapy demonstrate increased frequency of molecularly defined primitive cells at diagnosis. These results were validated by deconvolution of an independent previously published dataset of bulk transcriptomes from 59 CML patients. They further applied a BCR-ABL-associated gene signature to classify primitive Lin-CD34+CD38- stem cells as BCR:ABL+ and BCR:ABL-. They identified variability in the ratio of leukemic to non-leukemic primitive cells between patients, showed differences in the expression of cell surface markers, and determined that a combination of CD26 and CD35 cell surface markers could be used to prospectively isolate the two populations. The relative proportion of CD26-CD35+ (BCR:ABL-) primitive stem cells was higher in optimal responders compared to treatment failures, both at diagnosis and following 3 months of TKI therapy.

Strengths:

The studies are carefully conducted and the results are very clearly presented. The data generated will be a valuable resource for further studies. The strengths of this study are

the application of single-cell multi-omics using CITE-Seq to study individual variations in stem and progenitor clusters at diagnosis that are associated with good versus poor outcomes in response to TKI treatment. These results were confirmed by deconvolution of a historical bulk RNAseq data set. Moreover, they are also consistent with a recent report from Krishnan et al. and are a useful confirmation of those results. The major new contribution of this study is the use of gene expression profiles to distinguish BCRABL+ and BCR-ABL- populations within CML primitive stem cell clusters and then applying antibody-derived tag (ADT) data to define molecularly identified BCR:ABL+ and BCR-ABL- primitive cells by expression of surface markers. This approach allowed them to show an association between the ratio of BCR-ABL+ vs BCR-ABL- primitive cells and TKI response and study dynamic changes in these populations following short-term TKI treatment.

Weaknesses:

One of the limitations of the study is the small number of samples employed, which is insufficient to make associations with outcomes with confidence. Although the authors discuss the potential heterogeneity of primitive stem, they do not directly address the heterogeneity of hematopoietic potential or response to TKI treatment in the results presented. Another limitation is that the BCR-ABL + versus BCR-ABL- status of cells was not confirmed by direct sequencing for BCR-ABL. The BCR-ABL status of cells sorted based on CD26 and CD35 was evaluated in only two samples. We also note that the surface markers identified were previously reported by the same authors using different single-cell approaches, which limits the novelty of the findings. It will be important to determine whether the GEP and surface markers identified here are able to distinguish BCR-ABL+ and BCR-ABL- primitive stem cells later in the course of TKI treatment. Finally, although the authors do describe differential gene expression between CML and normal, BCR:ABL+ and BCR:ABL-, primitive stem cells they have not as yet taken the opportunity to use these findings to address questions regarding biological mechanisms related to CML LSC that impact on TKI response and outcomes.

Reviewer #1 (Recommendations For The Authors):

Minor comment: Fig 4 legend -E and F should be C and D.

We thank the reviewer for positive assessment of our work. Here, we highlight the updates in the revised manuscript considering the feedback received.

Minor comment: Fig 4 legend -E and F should be C and D.

We have edited the revised manuscript accordingly

One of the limitations of the study is the small number of samples employed, which is insufficient to make associations with outcomes with confidence.

Although we performed CITE-seq for 9 CML patient samples at diagnosis, we extended our investigations to include additional samples (e.g., largescale deconvolution analysis of samples, Fig 3 C-E, qPCR for BCR::ABL1 status, Fig. 6A, and the ratio between CD35+ and CD26+ populations at diagnosis and during TKI therapy, Fig. 6C-D) as described in the manuscript.

In comparison to a scRNA-seq, multiomic CITE-seq involves preparation and sequencing of separate libraries corresponding to RNA and ADTs thereby being even more resource demanding limiting our capacity to process an extensive number of patient samples. To confirm our findings in a larger cohort we have therefore adopted a computational deconvolution approach, CIBERSORT to analyze a larger number of independent samples (n=59). This reflects a growing, sustainable trend to study larger number of patients in face of

still prohibitively expensive but potentially insightful scomics approaches (For example, please see Zeng et al, A cellular hierarchy framework for understanding heterogeneity and predicting drug response in acute myeloid leukemia, Nature Medicine, 2022).

However, in view of the comment, we have now substantially increased the number of analyzed patients in the revised manuscript. These include increased number of patient samples to investigate the ratio between CD35 and CD26 marked populations at diagnosis, and 3 months of TKI therapy (from n=8 to n=12 with now 6 optimal responders and 5 treatment failure at diagnosis and after TKI therapy), qPCR for BCR::ABL1 expression status at diagnosis (from n=3 to n=9), and followed up the BCR::ABL1 expression in three additional samples after TKI therapy. Moreover, we examined the CD26 and CD35 marked populations for expression of GAS2, one of our top candidate LSC signature genes in three additional samples at diagnosis and at 3m follow up. Thus, >80 patient samples across different approaches have been analyzed to strengthen all major conclusions of the study.

We emphasize that we were cautious in generalizing the observation obtained from any one approach and sought to confirm any major finding using at least one complementary method. As an example, although CITE-seq (n=9) showed altered frequency of all cell clusters between optimal and poor responders (Fig. 3B), we refrained from generalizing because our independent large-scale computational deconvolution analysis (n=59) only substantiated the altered proportion of primitive and myeloid cell clusters (Fig. 3E).

Although the authors discuss the potential heterogeneity of primitive stem, they do not directly address the heterogeneity of hematopoietic potential or response to TKI treatment in the results presented.

Thanks for noting the discussion on heterogeneity of the primitive stem cells. As described in the original manuscript, the figure 6 D-E showed a relationship between heterogeneity and TKI therapy response. The results showed that CD35+/CD26+ ratio within the HSC fraction associated with this therapy response. We have now increased the number of patient samples analyzed and present the updated results in the revised manuscript (now figure 6 C-D). These observations set the stage for assessing whether long term therapy outcome can also be influenced by heterogeneity at diagnosis.

We have shown the hematopoietic potential of HSCs marked by CD35 expression in an independent parallel study and therefore only mentioned it concisely in the current manuscript. A combination of scRNA-seq, scATAC-seq and cell surface proteomics showed CD35+ cells at the apex of healthy human hematopoiesis, containing an HSCspecific epigenetic signature and molecular program, as well as possessing self-renewal capacity and multilineage reconstitution in vivo and vitro. The preprint is available as Sommarin et al. 'Single-cell multiomics reveals distinct cell states at the top of the human hematopoietic hierarchy', Biorxiv; <https://www.biorxiv.org/content/10.1101/2021.04.01.437998v2.full>

We also note that the surface markers identified were previously reported by the same authors using different single-cell approaches, which limits the novelty of the findings.

Our current manuscript is indeed a continuation of and builds onto our previous paper (Warfvinge R et al. Blood, 2017). In contrast to our previous report which was limited to examination of only 96 genes per cell, CITE-seq allowed us to examine the molecular program of cells using unbiased global gene expression profiling. Finally, although CD26 appears, once again as a reliable marker of BCR::ABL1+ primitive cells, CD35 emerges as a novel and previously undescribed marker of BCR::ABL1- residual stem cells. A combination of CD35 and CD26 allowed us to efficiently distinguish between the two populations housed within the Lin-34+38/low stem cell immunophenotype.

Another limitation is that the BCR-ABL + versus BCR-ABL- status of cells was not confirmed by direct sequencing for BCR-ABL. The BCR-ABL status of cells sorted based on CD26 and CD35 was evaluated in only two samples

Single cell detection of fusion transcripts is challenging with low detection sensitivity in single cell RNA-seq as has been noted previously (Krishnan et al. Blood, 2023, Giustacchini et al. Nature Medicine, 2017, Rodriguez-Meira et al. Molecular Cell, 2019). However, this is likely to change with the inclusion of targetspecific probes in scRNA-seq library preparation protocols. Nonetheless, in view of the comment, we have included more patient samples (from the previous n=3 to current n=10 (including TKI treated samples) for direct assessment of BCR-ABL1 status by qPCR analysis; the updated results are included in the revised manuscript (Figure 6A).

It will be important to determine whether the GEP and surface markers identified here are able to distinguish BCR-ABL+ and BCR-ABL- primitive stem cells later in the course of TKI treatment.

We performed qPCR to check for BCR::ABL1 status, and the level of GAS2, one of the top genes expressed in CML cells within CD26+ and CD35+ cells at diagnosis and following 3 months of TKI therapy. The results showed that while CD26+ are BCR::ABL1+, the CD35+ cells are BCR::ABL1- at both time points. Moreover, the expression of LSC-specific gene, GAS2 was specific to BCR::ABL1+ CD26+ cells at both diagnosis as well as following 3 months of TKI therapy. The new results are presented in figure 6B in the revised manuscript.

Finally, although the authors do describe differential gene expression between CML and normal, BCR:ABL+ and BCR:ABL-, primitive stem cells they have not as yet taken the opportunity to use these findings to address questions regarding biological mechanisms related to CML LSC that impact on TKI response and outcomes.

We agree with the reviewer that our major focus here was to characterize the cellular heterogeneity coupled to treatment outcome and therefore we did not delve deep into the molecular mechanisms underlying TKI response. However, in response to this comment, as mentioned above, we noted that one of the top genes in BCR::ABL1 cells (Fig. 4 C; right; in red), GAS2 (Growth Specific Arrest 2) was expressed at both diagnosis and TKI therapy within CD26+ cells relative to CD35+ cells (updated figure 6B). Interestingly, GAS2 was also detected in CML LSCs in a recent scRNA-seq study (Krishnan et al. Blood, 2023) suggesting GAS2 upregulation could be a consistent molecular feature of CML cells. GAS2 has been previously noted as deregulated in CML (Janssen JJ et al. Leukemia, 2005, Radich J et al, PNAS, 2006), control of cell cycle, apoptosis, and response to Imatinib (Zhou et al. PLoS One, 2014). Future investigations are warranted to assess whether GAS2 could play a role in the outcome of long-term TKI therapy.

Reviewer 2:

Summary:

The authors use single-cell "multi-comics" to study clonal heterogeneity in chronic myeloid leukemia (CML) and its impact on treatment response and resistance. Their main results suggest 1) Cell compartments and gene expression signatures both shared in CML cells (versus normal), yet 2) some heterogeneity of multiomic mapping correlated with ELN treatment response; 3) further definition of s unique combination of CD26 and CD35 surface markers associated with gene expression defined BCR::ABL1+ LSCs and BCR::ABL1- HSCs. The manuscript is well-written, and the method and figures are clear and informative. The results fit the expanding view of cancer and its therapy as a

complex Darwinian exercise of clonal heterogeneity and the selective pressures of treatments.

Strengths:

Cutting-edge technology by one of the expert groups of single-cell 'comics.

Weaknesses:

Very small sample sizes, without a validation set. The obvious main problem with the study is that an enormous amount of results and conjecture arise from a very small data set: only nine cases for the treatment response section (three in each of the ELN categories), only two normal marrows, and only two patient cases for the division kinetic studies. Thus, it is very difficult to know the "noise" in the system - the stability of clusters and gene expression and the normal variation one might expect, versus patterns that may be reproducibly study artifact, effects of gene expression from freezing-thawing, time on the bench, antibody labeling, etc. This is not so much a criticism as a statement of reality: these elegant experiments are difficult, timeconsuming, and very expensive. Thus in the Discussion, it would be helpful for the authors to just frankly lay out these limitations for the reader to consider. Also in the Discussion, it would be interesting for the authors to consider what's next: what type of validation would be needed to make these studies translatable to the clinic? Is there a clever way to use these data to design a faster/cheaper assay?

We thank the reviewer for appraisal of our manuscript. We take the opportunity to point out the updates in the revised manuscript in view of the comments.

Very small sample sizes, without a validation set. The obvious main problem with the study is that an enormous amount of results and conjecture arise from a very small data set: only nine cases for the treatment response section (three in each of the ELN categories), only two normal marrows, and only two patient cases for the division kinetic studies.

As the reviewer has noted the single cell omics experiments remain resource demanding thereby placing a limitation on the number of patients analyzed. As described above in response to the comments from reviewer 1, multiomic CITE-seq allows extraction of two modalities in comparison to a typical scRNA-seq, however, this also makes it even more limited in the number of samples processed in a sustainable way. This was one of the motivations to analyze a larger number of independent samples (n=59) while benefiting from the insights gained from CITE-seq (n=9). Furthermore, by analyzing CD34+ cells from bone marrow and peripheral blood of CML patients, including both responders and non-responders after one year of Imatinib therapy, we were able to significantly diversity the patient pool, which was lacking in our CITE-seq patient pool. As mentioned above, this reflects a growing trend to analyze larger number of patients while anchoring the analysis on prohibitively expensive but potentially insightful sc-omics approaches (For example, please see Zeng et al, A cellular hierarchy framework for understanding heterogeneity and predicting drug response in acute myeloid leukemia, Nature Medicine, 2022).

As emphasized above, we frequently sought to confirm the findings from one approach using a complementary method and independent samples. For example, although CITE-seq (n=9) showed altered frequency of all cell clusters between optimal and poor responders (Fig. 3B), we refrained from generalizing because an independent largescale computational deconvolution analysis (n=59) only substantiated the altered proportion of primitive and myeloid clusters.

In view of the comment, we have now increased the number of patients analyzed during the revision process. These include increased numbers to investigate the ratio between CD35+ and CD26+ populations at diagnosis, as well as 3 months of TKI therapy, qPCR for BCR::ABL1, and patients examined for GAS2, one of the top genes expressed in CML cells (see response to reviewer 1 for details). Altogether, >80 patient samples across different approaches were analyzed to strengthen the conclusions.

During the revision, we have analyzed cells from 8 CML patients for cell cycle using gene activity scores. This is in addition to the cell division kinetics data reported previously are now together described in the supplementary figures 9C-F.

It is very difficult to know the "noise" in the system - the stability of clusters and gene expression and the normal variation one might expect, versus patterns that may be reproducibly study artifact, effects of gene expression from freezing-thawing, time on the bench, antibody labeling, etc. This is not so much a criticism as a statement of reality: these elegant experiments are difficult, time-consuming, and very expensive. Thus in the Discussion, it would be helpful for the authors to just frankly lay out these limitations for the reader to consider.

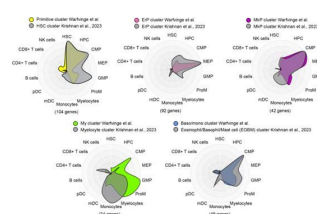
We agree with the reviewer that sc-omics approaches can be noisy despite continuing efforts to denoise single cell datasets through both experimental and bioinformatic innovations. Therefore, we have updated the discussion as recommended by the reviewer (paragraph 5 in the discussion).

We also note that CITE-seq, in contrast to scRNA-seq alone provides dual features: surface marker/protein as well as RNA for annotating the same cluster. In our manuscript, for example, cell clusters in UMAP for normal BM; Fig 1B were described using both surface markers (Fig. 1C) and RNA (Fig. 1D) making the cluster identity robust. To further elaborate this approach, a new supplementary figure 1C shows annotations of clusters using both RNA and surface markers.

To potentially address the issue of stability of clusters and gene expression, we compared the marker genes for major clusters from nBM from this study (supplementary table 4, Warfvinge et al.) with those described recently in a scRNA-seq study by Krishnan et al. supplementary table 8, Blood, 2023 using Cell Radar, a tool that identifies and visualizes which hematopoietic cell types are enriched within a given gene set (description: <https://github.com/KarlssonG/cellradar>

Direct link: <https://karlssong.github.io/cellradar/>). To compare, we used our in-house gene list for the major clusters as well as mapped the same number of top marker genes based on log2FC from corresponding cluster from Krishnan et al. as inputs to Cell Radar. The Cell Radar plot outputs are shown below.

Author response image 1.



This approach showed broad similarities across clusters from this study with their counterparts from the other study suggesting the cluster identities reported here are likely to

be robust. Please note these figures are for reviewer response only and not included in the final manuscript.

Also in the Discussion, it would be interesting for the authors to consider what's next: what type of validation would be needed to make these studies translatable to the clinic? Is there a clever way to use these data to design a faster/cheaper assay?

Our findings on CD26+ and CD35+ surface markers to enrich BCR::ABL1+ and BCR::ABL1- cells suggest a simpler, faster and cheaper FACS panel can possibly quantify leukemic and non-leukemic stem cells in CML patients. We anticipate that future investigations, clinical studies might examine whether CD26CD35+ cells could be plausible candidates for restoring normal hematopoiesis once the TKI therapy diminishes the leukemic load, and whether patients with low counts of CD35+ cells at diagnosis have a relatively higher chance of developing hematologic toxicity such as cytopenia during therapy.

We briefly mentioned this possibility in the discussion; however, we have now moved it to another paragraph to highlight the same. Please see paragraph 5 in the revised manuscript.

Reviewer 3:

Summary:

In this study, Warfvinge and colleagues use CITE-seq to interrogate how CML stem cells change between diagnosis and after one year of TKI therapy. This provides important insight into why some CML patients are "optimal responders" to TKI therapy while others experience treatment failure. CITE-seq in CML patients revealed several important findings. First, substantial cellular heterogeneity was observed at diagnosis, suggesting that this is a hallmark of CML. Further, patients who experienced treatment failure demonstrated increased numbers of primitive cells at diagnosis compared to optimal responders. This finding was validated in a bulk gene expression dataset from 59 CML patients, in which it was shown that the proportion of primitive cells versus lineage-primed cells correlates to treatment outcome. Even more importantly, because CITE-seq quantifies cell surface protein in addition to gene expression data, the authors were able to identify that BCR/ABL+ and BCR/ABL- CML stem cells express distinct cell surface markers (CD26+/CD35- and CD26-/CD35+, respectively). In optimal responders, BCR/ABL- CD26-/CD35+ CML stem cells were predominant, while the opposite was true in patients with treatment failure. Together, these findings represent a critical step forward for the CML field and may allow more informed development of CML therapies, as well as the ability to predict patient outcomes prior to treatment.

Strengths:

This is an important, beautifully written, well-referenced study that represents a fundamental advance in the CML field. The data are clean and compelling, demonstrating convincingly that optimal responders and patients with treatment failure display significant differences in the proportion of primitive cells at diagnosis, and the ratio of BCR-ABL+ versus negative LSCs. The finding that BCR/ABL+ versus negative LSCs display distinct surface markers is also key and will allow for a more detailed interrogation of these cell populations at a molecular level.

Weaknesses:

CITE-seq was performed in only 9 CML patient samples and 2 healthy donors. Additional samples would greatly strengthen the very interesting and notable findings.

Reviewer #3 (Recommendations For The Authors):

My only recommendation is to bolster findings with additional CML and healthy donor samples.

CITE-seq was performed in only 9 CML patient samples and 2 healthy donors. Additional samples would greatly strengthen the very interesting and notable findings.

We thank the reviewer for the positive assessment of our manuscript. As mentioned in response to comments from reviewer 1 and 2, CITE-seq remains an resource consuming single cell method potentially limiting the number of patients to be analyzed. However, during the revision process, we have increased the number of patient material analyzed for other assays; these include increased number to investigate the ratio between CD35+ and CD26+ populations at diagnosis, and 3 months of TKI therapy, qPCR for BCR::ABL1, and patients examined for GAS2, one of the top genes expressed in CML cells. Thus, >80 patient samples across different assays have been analyzed to strengthen the conclusions. (Please see comment to reviewer 1 for more details)